

Global risk mapping of highly pathogenic avian influenza H5N1 and H5Nx in the light of epidemic episodes occurring from 2020 onward

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This global study compares environmental niche model outputs of avian influenza pathogen niche constructed for two distinct periods, and uses differences between those outputs to suggest that the changed case numbers and distribution relate to intensification of chicken and duck farming, and extensive cultivation. While a **useful** update to existing niche models of highly pathogenic avian influenza, the justification for the use of environmental niche models to explore correlative relationships between land cover change and changed case epidemiology is **incomplete**. Key assumptions have not been adequately clarified for the readers benefit, and in consequence the communication of the likely limitations of the work are not sufficiently clear.

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Abstract

Avian influenza (AI) is a highly contagious viral disease affecting poultry and wild water birds, posing significant global challenges due to its high mortality rates and economic impacts. Highly pathogenic avian influenza (HPAI) outbreaks, particularly those caused by H5N1 and its variants, have surged since their first occurrence in 1959. The HPAI H5N1 clade 2.3.4.4b viruses have notably expanded their geographical reach, affecting numerous countries, diverse avian species, and now wild and domestic mammals. Using an ecological niche modelling approach, this study aims to elucidate the environmental factors associated with the increased HPAI H5 cases since 2020, investigate potential shifts in ecological niches, and predict new areas suitable for local viral circulation. Focusing on H5N1 and H5Nx

strains, we have developed ecological niche models for HPAI cases in both wild and domestic birds while considering two distinct periods: 2015-2020 and 2020-2022. Key environmental predictors include chicken and duck population density, human density, distance to water bodies, and several land cover variables. Post-2020, we observe a notable increase in the relative importance of some of these predictors, such as intensive chicken population density and cultivated vegetation. The resulting risk maps reveal notable ecological suitability for local HPAI H5 circulation in Europe, Asia, as well as North and South America, with notable expansions of the areas at risk post-2020. The spatial distribution of HPAI H5 occurrences in wild birds appears to be primarily correlated with urban areas and open water regions. Overall, global risk maps derived from our models identify regions at risk where surveillance and control measures should be prioritised. Finally, our analyses also highlight a potential shift in the diversity of wild bird species affected by HPAI outbreaks, with a higher variety of avian species, particularly sea birds, being impacted post-2020. This increased diversity could suggest that ecological shifts in HPAI H5 circulation may be accompanied by a broader range of susceptible species. Overall, these results further contribute to the understanding of HPAI epidemiology.

Introduction

Since 2020, an increase in both H5Nx and H5N1 cases has been observed (**Fig. 1** [↗](#)). A variant of the Gs/Gd H5N1 viruses belonging to the H5 clade 2.3.4.4b has led to an unprecedented number of deaths in wild birds and poultry in many countries in Africa, Asia, and Europe ([Huang et al., 2023](#) [↗](#)). In 2021, the virus spread to North America, and in 2022, to Central and South America ([FAO, WHO and WOA, 2023](#) [↗](#)) spilling over into poultry farms and infecting an alarming number of wild terrestrial, marine and domestic mammals ([Elsmo et al., 2023](#) [↗](#); [Leguia et al., 2023](#) [↗](#); [Neumann and Kawaoka, 2024](#) [↗](#); [Peacock et al., 2024](#) [↗](#); [Plaza et al., 2024](#) [↗](#)); marking an unprecedented expansion in the geography and impact of HPAI. In 2022, 67 countries across five continents reported H5N1 HPAI outbreaks in poultry and wild birds to the World Organisation for Animal Health (WOAH), resulting in over 131 million domestic poultry deaths or cullings ([WOAH, 2023](#) [↗](#)). In 2023, another 14 countries, mostly in the Americas, reported outbreaks. Several mass death events in wild birds were caused by influenza A(H5N1) clade 2.3.4.4b viruses ([Klaassen and Wille, 2023](#) [↗](#); [WOAH, 2023](#) [↗](#)).

In that epidemiological context, understanding the environmental factors associated with the risk of HPAI circulation remains crucial. Extensive exploration of risk factors for HPAI presence, spread, and persistence has been conducted in various countries and regions. These studies have identified domestic waterfowl, several anthropogenic variables (human population density, distance to roads) and indicators of water presence as important factors for risk of H5N1 local circulation ([Gilbert and Pfeiffer, 2012](#) [↗](#)). More recent work on global suitability for HPAI have pinpointed that host-related variables such as poultry density are the strongest contributors to HPAI persistence ([Dhingra et al., 2016](#) [↗](#); [Gierak and Śmietanka, 2021](#) [↗](#); [Martin et al., 2011](#) [↗](#)).

In addition, poultry intensification, international poultry trade, live bird markets, and wild bird migratory routes have been recognised as playing a key role in the transmission and spread of HPAI ([Vandegrift et al., 2010](#) [↗](#)). Lastly, evidence of climate change impacting the dynamics of HPAI has also been discussed ([Prosser et al., 2023](#) [↗](#)). For example, changes in rainfall alter the distribution, abundance, and quality of wetlands can impact waterfowl populations ([Forcey et al., 2007](#) [↗](#); [Gilbert et al., 2008](#) [↗](#); [Vandegrift et al., 2010](#) [↗](#)). However, recent outbreaks raise questions about whether earlier ecological niche models still accurately predict the current distribution of areas ecologically suitable for the local circulation of HPAI H5 viruses.

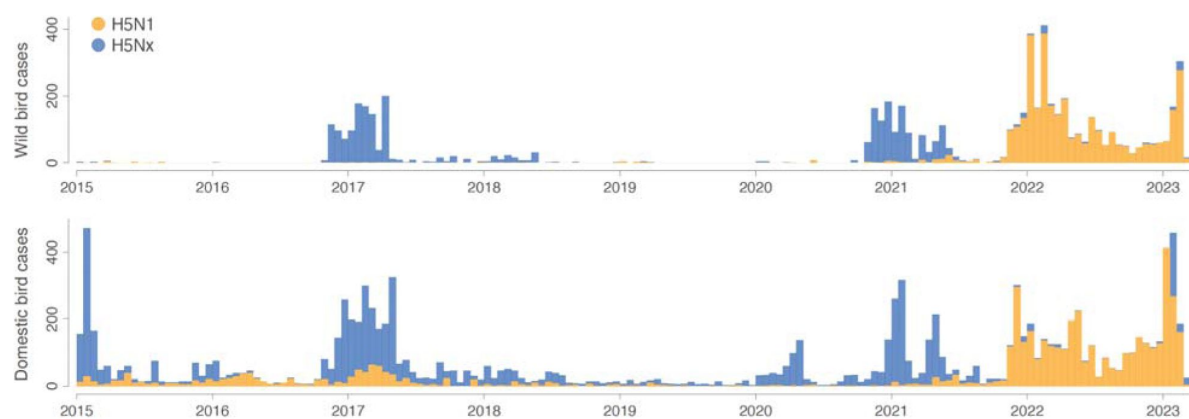


Figure 1

Epidemic curves for both wild and domestic bird cases.

Orange and blue histograms report the weekly number of H5N1 cases and the weekly number of H5Nx cases, respectively.

Understanding the interplay between environmental variables and avian influenza becomes increasingly crucial to have an in-depth understanding of the risk factors that govern avian influenza viruses (AIVs) circulation and spread. Following a previous study dedicated to the ecological niche of H5Nx and H5N1 (Dhingra et al., 2016 [↗](#)), we aimed to address the following questions: (i) Which factors can explain the observed increase in H5Nx and H5N1 cases since 2020? (ii) Do we observe a change or an extension of the ecological conditions suitable for local HPAI circulation? (iii) Are we able to predict the areas at risk for local HPAI circulation based on ecological niche models trained on occurrence data before 2020? To this end, we used HPAI H5N1 and H5Nx occurrence data collected before 2020 to train ecological niche models and tested whether these models could accurately predict areas ecologically suitable for HPAI circulation post-2020. We developed separate ecological niche models for wild and domestic bird HPAI occurrences, these models thus predicting the ecological suitability for the risk of local viral circulation leading to the detection of HPAI occurrences within each host group (rather than the niche of the virus or the host species alone). Specifically, we employed a boosted regression trees (BRT) approach to estimate the HPAI ecological suitability given local environmental conditions. We also computed the diversity indices of wild bird species involved in HPAI occurrences, helping to understand how shifts in species diversity might correspond with the environmental factors considered in the ecological niche modelling analyses of HPAI in wild or domestic birds. Our results provide a better understanding of HPAI dynamics by identifying key environmental factors correlated with the increase in H5Nx and H5N1 cases in poultry and wild birds, investigating potential shifts in their ecological niches, and improving the prediction of at-risk areas. This knowledge might aid policymakers, epidemiologists, and wildlife conservationists in better targeting surveillance and control measures.

Results

In this study, we applied an ecological niche modelling approach using the boosted regression trees (BRT) method to evaluate the distribution of areas ecologically suitable for the risk of local circulation of HPAI H5Nx and H5N1 viruses in wild and domestic bird populations across two periods: 2015-2020 and 2020-2022. We trained separate ecological niche models for each combination of host species (wild or domestic birds), virus strain (H5Nx or H5N1), and time period. In addition to evaluating each model separately, models trained on data from the 2015-2020 period were tested to predict occurrences in the 2020-2022 period, allowing us to evaluate the potential of pre-2020 models to predict a more recent distribution of H5N1 and H5Nx cases.

We assessed the influence of a broad range of spatial predictors — including anthropogenic, topographical, land cover, eco-climatic, and poultry density — on virus occurrences (**Fig. S1** [↗](#)). We used occurrence records from the EMPRES-i database and sampled pseudo-absence points to account for the lack of virus detection data in some areas. These pseudo-absence points were distributed based on human population density, with more pseudo-absence points sampled in areas of higher population to reflect the greater surveillance efforts in those regions (**Fig. S2** [↗](#)).

The accuracy of the models was validated through three types of cross-validation: (i) a standard cross-validation with a random and stratified divide between training and validation sets, (ii) a spatial cross-validation based on the approach used by Dhingra and colleagues (2016) [↗](#) that clustered presence and pseudo-absence points into folds using reference presence points, and (iii) a second spatial cross-validation based on a blocks generation technique (Valavi et al., 2019 [↗](#)). Models using spatial cross-validation techniques are less affected by spatial autocorrelation, as shown by higher spatial sorting bias (SSB) values (**Fig. S3** [↗](#)). Among the seven datasets evaluated, four show better performance with the reference points-based spatial cross-validation method, suggesting it was less impacted by spatial autocorrelation compared to the block generation technique. Consequently, we have proceeded with the reference points approach for consistency with Dhingra et al. (2016) [↗](#).

Using pre-2020 data, our models demonstrated a relatively robust capability to predict the distribution of post-2020 occurrence data. The Area Under the Curve (AUC) values for ecological niche models trained on occurrence data in domestic birds before 2020 ranged between 0.74 and 0.77 when evaluated against post-2020 data, indicating a relatively good predictive performance. In comparison, ecological niche models trained on post-2020 data showed a slightly higher predictive performance when evaluated on data from the same period, with AUC values ranging from 0.78 to 0.83 (**Table S1**). These findings underscore the ability of pre-2020 models in forecasting the recent geographic distribution of ecological suitability for H5Nx and H5N1 occurrences.

Previous literature reviews (Gilbert and Pfeiffer, 2012; Dhingra et al., 2016) have summarised the predictor variables commonly correlated with HPAI occurrences. As detailed in the Materials and Methods section and in the table in Supplementary Information Resources S1, we explored four sets of environmental variables considered by Dhingra and colleagues (Dhingra et al., 2016): host variables (set 1), land cover variables (set 2), eco-climatic variables (set 3), and a risk-based selection of variables performed by Dhingra and colleagues (set 4). Set 3 showed the lowest predictive performance for both domestic and wild bird cases during the two periods and, in contrast, sets 2 and 4 demonstrated the highest predictive accuracy for wild and domestic birds, respectively (**Fig. S3**). Set 4 combined host variables with cultivated and managed vegetation, open water areas, distance to water, and the annual mean of land surface temperature. For domestic birds, set 1 (host variables) performed well before 2020, but its predictive performance decreased after 2020 in favour of set 4. Ecological niche models trained on wild and domestic bird cases, respectively with the sets 2 and 4, are associated with AUC values exceeding 0.77, indicating a relatively strong performance in predicting H5 occurrences based on these sets of environmental variables. **Figure 2** displays the response curves of the most important variables with their respective relative influence (RI) (**Table S2**). In the ecological niche models for H5N1 and H5Nx in domestic birds, the densities of intensive chicken populations, domestic duck populations, and human populations emerged as significant predictors, each with RI values exceeding 5%. Notably, the RI for intensive chicken density in H5N1 models increased sharply from 8.5% to 30.4% since 2020. Similarly, the RI of cultivated and managed vegetation has doubled for both strains post-2020. The response curves showed a positive correlation, indicating that higher values of these predictors were linked with an increased likelihood of HPAI occurrences given local environmental conditions. These findings indicate a trend towards increased HPAI susceptibility in environments characterised by intensive agricultural and vegetation management practices.

In contrast, eco-climatic factors such as land surface temperature and precipitation (set 3) showed only moderate influence. We also observed a decrease in the importance of domestic duck population density for both subtypes after 2020, possibly due to the increasing diversity of bird species involved in the transmission dynamics of H5N1 and H5NX (**Fig. S4**). These findings highlight the crucial role of anthropogenic and host-related variables in accurately predicting HPAI occurrences.

In the ecological niche models trained for wild bird cases, urban and built-up areas were associated with H5N1 and H5Nx occurrences (**Fig. 2**), showing the highest relative importance (RI) prior to 2020. Specifically, RI values reached 54.5% before 2020 but decreased to 39.3% in the post-2020 period. This decline may indicate a reduced bias in observation data: typically, dead wild birds are more frequently found near human-populated areas due to opportunistic detections, whereas more recent surveillance efforts have become increasingly proactive (Giacinti et al., 2024). After 2020, we observed an increase in the importance of habitats, such as deciduous broadleaf trees (6.1%), mixed and other tree regions (11.4%), and herbaceous vegetation for H5Nx (9.5%). However, response curves (**Fig. 2**) indicated that H5N1 and H5Nx occurrences in wild birds decreased in areas with greater tree cover, while it increased in regions dominated by

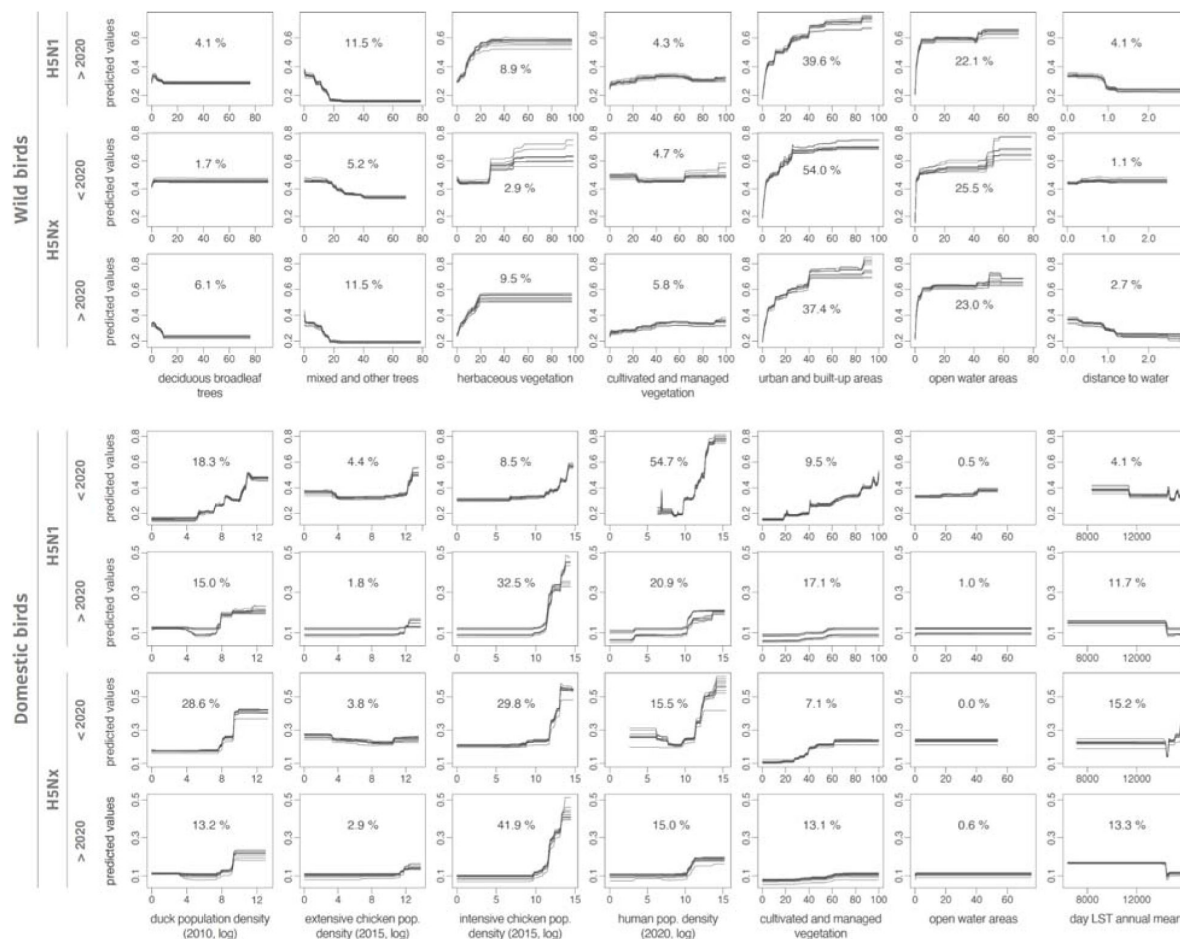


Figure 2

Response curves associated with the environmental variables included in the ecological niche models.

For the ecological niche models trained on wild bird infection records, we here only display the response curves estimated for the environmental variables associated with an averaged relative influence (RI) >4% for at least one of the considered occurrence datasets (thus not reporting the response curves obtained for the following variables: evergreen deciduous needleleaf trees, evergreen broadleaf trees, shrublands, and regularly flooded vegetation). Each curve was retrieved from a distinct boosted regression tree (BRT) model trained for a specific dataset of occurrence data. We also report the averaged RI (in %) of each environmental variable in the respective ecological models trained on a specific dataset of occurrence data (see [Table S1](#) for the complete list of RI estimates along with their first and third quartiles). Due to lack of data, the model was not trained for H5N1 in wild birds before 2020.

herbaceous vegetation. Open water areas consistently showed high RI values across all time periods and virus strains, particularly for H5Nx before 2020 (25.5%) and H5N1 after 2020 (22.0%), highlighting their significant role in the ecological models of HPAI outbreaks.

Figure 3 displays maps of the predicted ecological suitability for the risk of local H5N1 and H5Nx circulation across two time periods, 2015–2020 and 2020–2022; these maps being also accessible on the following link for a dynamic visualisation of the results: <https://app.mood-h2020.eu/core>. All maps are in a WGS84 projection with a spatial resolution of 0.0833 decimal degrees (i.e. 5 arcmin, or approximately 10 km at the equator). For H5N1 and H5Nx in domestic birds, these maps reveal several regions associated with a relatively high ecological suitability, especially in Europe and Asia, in countries such as South Korea, Japan, Singapore, Malaysia, Vietnam, Cambodia, Thailand, and the Philippines, as well as the United Kingdom, France, the Netherlands, Germany, Italy, Ukraine, and Poland. Additionally, several areas in African nations, including Nigeria and South Africa, were identified as suitable environments for both H5N1 and H5Nx local circulation in scenarios before and after 2020. Regions in North America and South America — including Bolivia, Brazil, Colombia, Ecuador, Chile, Peru, and Venezuela — demonstrated ecological suitability for the risk of local circulation for H5Nx and H5N1, with a marked increase in the predicted suitability for H5N1 after 2020. For H5Nx post-2020, areas of high predicted ecological suitability, such as Brazil, Bolivia, the Caribbean islands, and Jilin province in China, likely result from extrapolations, as these regions reported few or no outbreaks in the training data.

Our findings were overall consistent with those previously reported by Dhingra and colleagues (Dhingra et al., 2016), who used data from January 2004 to March 2015 for domestic poultry. However, some differences were noted: their maps identified higher ecological suitability for H5 occurrences before 2016 in North America, West Africa, eastern Europe, and Bangladesh, while our maps mainly highlight ecologically suitable regions in China, South-East Asia, and Europe (Fig. S5). In India, analyses consistently identified high ecologically suitable areas for the risk of local H5Nx and H5N1 circulation for the three time periods (pre-2016, 2016–2020, and post-2020). Similar to the results reported by Dhingra and colleagues, we observed an increase in the ecological suitability estimated for H5N1 occurrence in South America's domestic bird populations post-2020. Finally, Dhingra and colleagues identified high suitability areas for H5Nx occurrence in North America, which are predicted to be associated with a low ecological suitability in the 2016–2020 models.

For H5Nx and H5N1 cases in wild birds, the estimated ecological niches were primarily associated with environmental factors such as open water and distance to water, which result in ecological suitability hotspots estimated near coasts and main rivers. For both H5Nx and H5N1, certain areas of predicted high ecological suitability appear spatially isolated, i.e. surrounded by regions of low predicted ecological suitability. These areas likely meet the environmental conditions associated with past HPAI occurrences, but their spatial isolation may imply a lower risk of actual occurrences, particularly in the absence of nearby outbreaks or relevant wild bird movements. Some of the areas with high predicted ecological suitability reflect the result of extrapolations. This is particularly the case in coastal regions of West and North Africa, the Nile Basin, Central Asia (Kyrgyzstan, Tajikistan, Uzbekistan), Brazil (including the Amazon and coastal areas), southern Australia, and the Caribbean, where ecological conditions are similar to those in areas where outbreaks are known to occur but where records of outbreaks are still rare.

We observed that areas of relatively high ecological suitability have expanded after 2020. North America, particularly near the Great Lakes region, appeared more suitable for local H5Nx circulation. This expansion was also evident in Russia and South America, aligning with major bird migration routes. Notably, while H5Nx had not been reported in Australia, the models indicate potential ecological suitability there as well. Expanded ecological suitability was also

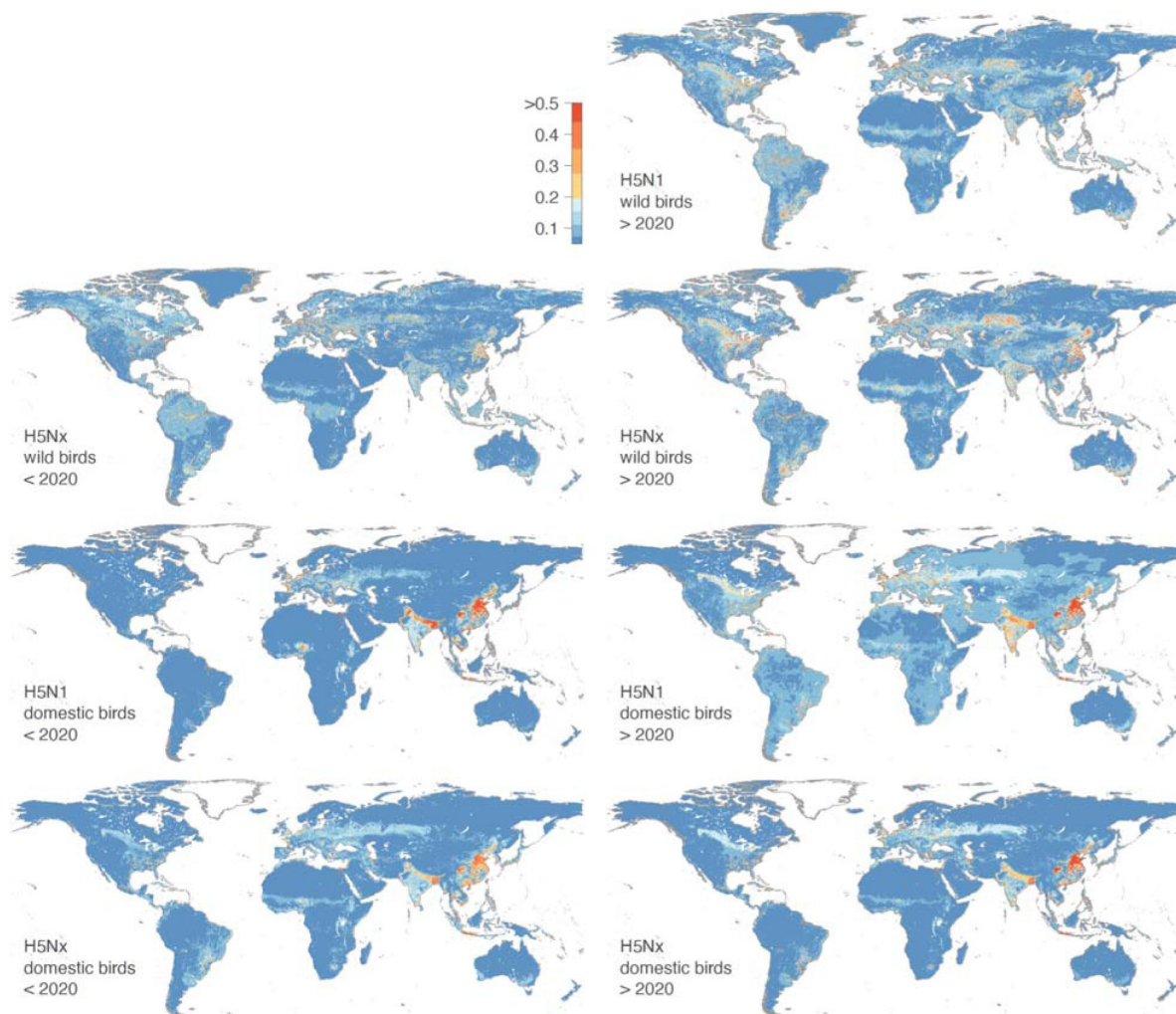


Figure 3

Areas ecologically suitable for local H5N1 or H5Nx circulation leading to infection cases in domestic bird populations.

We estimated the ecological suitability for two different time periods (2015-2020 and 2020-2022) and for both wild and domestic bird populations. Dynamic visualisations of the results are available here: <https://mood-platform.avia-gis.com/core>.

observed in China, India, and European countries. However, models trained on pre-2020 data maintained reasonable predictive performance when tested on post-2020 data, suggesting that the overall ecological niche of HPAI did not drastically shift over time.

In **Table 1**, we report the estimation of avian species diversity indices for species involved in HPAI outbreaks for the pre-2020 and post-2020 periods. Note that these indices reflect the diversity of bird species detected in outbreak records, not necessarily their abundance in the wild. We observed variations between these two periods both in the overall bird population and within specific wild bird species groups, including sea bird species (e.g., gulls, terns, boobies, gannets). For all birds, the Shannon index increased from 4.23 before 2020 to 5.03 after 2020, which might indicate a more diverse infected bird population in the latter period. The Simpson index however only rose from 0.97 to 0.98, but which could potentially suggest a slightly higher concentration of certain species post-2020. Sea birds exhibited a similar upward trend in diversity post-2020, with the Shannon index increasing from 2.19 to 2.26. This latter trend could potentially suggest a broader diversity of sea bird species involved in HPAI outbreaks.

When considering wild birds affected by the H5N1 strain of HPAI, there is a notable increase in the Shannon index from 2.00 before 2020 to 2.95 after 2020, accompanied by a rise in the Simpson index from 0.83 to 0.91. These increases could be indicative of greater diversity and a more even distribution of wild bird species among the reported HPAI H5N1 wild bird cases after 2020. In contrast, birds not affected by the H5N1 strain showed a slight decrease in diversity post-2020. The Shannon index decreases from 3.42 to 3.26, and the Simpson index decreases from 0.95 to 0.92. Given that HPAI particularly affected sea birds >2020, we further explore the yearly distribution of sea bird families. As depicted in **Figure S4**, positive cases among sea birds belonging to the families Laridae (gulls and terns), Sulidae (gannets and boobies), Ardeidae (herons), Pelecanidae (pelicans), Spheniscidae (penguins), Alcidae (auks), Phalacrocoracidae (cormorants) were detected more frequently from 2020 in comparison to before 2020. Notably, the occurrence of Laridae and Sulidae species among positive cases increased markedly from 2021 onwards.

To evaluate whether the post-2020 increase in species diversity estimated for infected wild birds could result from an increase in the number of tests performed on wild birds, we compared European annual surveillance test counts (EFSA et al., 2025, 2019) before and after 2020 using a Wilcoxon rank-sum test. We relied on European data because it was readily accessible and offered standardised and systematically collected metrics across multiple years, making it suitable for a comparative analysis. Although borderline significant (p -value = 0.063), the Wilcoxon rank-sum test indeed highlighted a recent increase in the number of wild bird tests (on average >11,000/year pre-2020 and >22,000 post-2020), which indicates that the comparison of bird species diversity metrics should be interpreted with caution. However, such an increase in the number of tests conducted in the context of a passive surveillance framework would thus also be in line with an increase in the number of wild birds found dead and thus tested. Therefore, while the increase in the number of tests could indeed impact species diversity metrics such as the Shannon index, it can also reflect an absolute higher wild bird mortality in line with a broadened range of infected bird species.

Discussion

Using an ecological niche modelling approach, our study presents an in-depth analysis of the ecological suitability — interpreted here as the risk of local circulation — for H5Nx and H5N1 HPAI viruses across different temporal and ecological contexts. By conducting boosted regression tree (BRT) analyses and incorporating a diverse set of spatial predictors, our study provides insights into the environmental and anthropogenic factors influencing the distribution of these viruses in both wild and domestic bird populations. For domestic birds, our results concur with those previously reported by Dhingra and colleagues (Dhingra et al., 2016), indicating that

	All birds		Sea birds		Wild birds		Domestic birds		All birds H5N1		All birds non-H5N1	
	<i>Shannon</i>	<i>Simpson</i>	<i>Shannon</i>	<i>Simpson</i>	<i>Shannon</i>	<i>Simpson</i>	<i>Shannon</i>	<i>Simpson</i>	<i>Shannon</i>	<i>Simpson</i>	<i>Shannon</i>	<i>Simpson</i>
< 2020	4.227	0.971	2.187	0.843	3.072	0.832	3.919	0.966	2.003	0.833	3.415	0.945
> 2020	5.033	0.982	2.255	0.831	5.066	0.983	2.967	0.897	2.952	0.907	3.254	0.914

Table 1.

Species diversity indices estimated from infected bird cases before and after 2020.

This table presents the Shannon and Simpson species diversity indices for various bird groups, comparing values before and after the year 2020. The indices are provided for all birds, sea birds, wild birds, domestic birds, as well as all birds affected by H5N1 and non-H5N1 strains.

ecological niche models incorporating host variables (chicken and duck population densities) outperform those based on land-use or eco-climatic factors in terms of predictive performance. Contrariwise, in our analysis of H5N1 and H5Nx HPAI in wild birds, the most significant predictors were anthropogenic factors, notably urban areas and distance to open water. However, observation biases might affect these findings since outbreaks in wild birds are commonly reported near human populations.

In our analyses, intensive chicken population density emerges as a significant predictor, underscoring the influence of poultry farming practices on HPAI spread (Dhingra et al., 2016 [↗](#); Vincenti-Gonzalez et al., 2024 [↗](#)) and revealing a complex interplay between ecological niches and the anthropogenic landscape. Similarly to the results of Dhingra and colleagues (Dhingra et al., 2016 [↗](#)), land-use and climatic predictors do not play an important role in the niche ecological models, even for wild birds. This suggests that HPAI H5Nx is not as strongly environmentally constrained as vector-borne pathogens, for which clear eco-climatic boundaries (e.g., vector borne diseases) can be mapped (Dhingra et al., 2016 [↗](#)). Additionally, we found that while domestic duck population density is a significant predictor for HPAI occurrence in domestic birds before 2020, its importance has decreased post-2020, particularly within the H5Nx ecological niche model. This observation is particularly noteworthy, as domestic ducks have historically been closely linked to areas of persistence and evolution of H5N1 HPAI (Gilbert and Pfeiffer, 2012 [↗](#)). However, our findings, in line with earlier studies such as the one of Dhingra and colleagues (Dhingra et al., 2016 [↗](#)), highlight a significant shift in the epidemiological dynamics of HPAI. Specifically, while the association between the H5Nx clade 2.3.4.4 and domestic ducks has for instance been documented in South Korea (Hill et al., 2015), it is becoming increasingly evident that HPAI is now more strongly associated with high-density chicken farming and anthropogenic factors. This shift suggests that the virus could now be circulating more among farms (farm to farm transmission) (FAO, 2023 [↗](#)), with fewer introductions by wild birds — particularly wild ducks, which have previously acted as “Trojan horses” for H5N1 HPAI (Kim et al., 2014). However, this remains an interpretation, as the available data do not allow us to distinguish between index cases and secondary transmission events. Consequently, we could be witnessing a shift from the traditional duck/rice ecosystems towards a scenario where intensive chicken farming and a wider diversity of wild birds play a more prominent role in the transmission dynamics of HPAI. Therefore, monitoring areas with high intensive chicken densities and regular surveillance of wild birds remains crucial for the early detection and management of HPAI outbreaks.

A difference between the ecological niche models previously trained by Dhingra and colleagues (Dhingra et al., 2016 [↗](#)) and our models is the pronounced increase in ecological suitability for H5N1 in domestic bird populations estimated in India and China. More generally, our models highlight the extensive spread of H5N1 and H5Nx HPAI among domestic birds in Asia previously identified as “reassortment sink areas” (Dhingra et al., 2018 [↗](#)). Reassortment refers to the process where influenza viruses exchange genetic material, potentially leading to new, more virulent strains. The extent of virus suitability in Asia highlights a potential change in the dynamics of virus spread, possibly driven by intensified poultry farming practices and increased human-wildlife interactions in these regions (Bahl et al., 2016 [↗](#); Gilbert et al., 2017 [↗](#)).

The recent outbreaks of infection in cattle within the United States have been partially delineated by our HPAI risk mapping approach, highlighting major hotspots predominantly in the Great Lakes region, including Michigan. From March to July 2024, approximately 26 herds in Michigan were affected (U.S. Department of Agriculture, 2024 [↗](#)). Additionally, our risk maps identify localised hotspots around urban centres in Colorado and Texas, where herds have also been infected. These outbreaks appear to stem from a single introduction event in Texas by wild birds (Worobey et al., 2024a [↗](#), 2024b [↗](#)). This trend may be indicative of an increasing risk of future introductions from wild birds, as we observed a significant rise in the diversity of infected bird species after 2020, particularly among sea birds.

The latest strain of H5N1 (H5 clade 2.3.4.4b), believed to have emerged in 2014, shows adaptations for infecting a wide range of wild birds and mammals (Caliendo et al., 2022 [↗](#); Graziosi et al., 2024 [↗](#)). By 2020, its spread among wild birds was three times faster than in farmed poultry due to mutations enabling the virus to infect diverse species (CDC, 2024; Xie et al., 2023). Unlike previous epidemic waves in Europe, outbreaks in the last years have also continued through summer, confirming that the virus's current circulation is no longer seasonal but nesting in the wild bird population (Sciensano, 2024 [↗](#)). Therefore, the virus is able to infect a wider range of species than previous forms, as it circulates all seasons (Sacristán et al., 2024 [↗](#)). The densely packed colonies of sea birds have also facilitated the rapid virus spread, resulting in high mortality in these sea bird colonies (Boulinier, 2023 [↗](#); Bregnballe et al., 2024 [↗](#)). For instance, the Netherlands lost up to 80% of its Sandwich terns in weeks, and the UK saw significant outbreaks, especially on Scottish islands (Knief et al., 2024 [↗](#)). In South America, similar trends have been observed, with notable die-offs in seabird populations along coastal regions, particularly impacting species such as penguins, pelicans and cormorants (Leguia et al., 2023 [↗](#)). Since then, in early 2023, the Peruvian outbreak had spread to marine mammals, particularly affecting the South American sea lion, which also began to experience a mass die-off (Campagna et al., 2024 [↗](#); Leguia et al., 2023 [↗](#)).

The increase in the number of wild bird species being infected by HPAI, particularly H5N1, could be attributed to several factors such as viral evolution, increased interaction between domestic and wild birds, climate and land-use changes, among others. Regarding land-use changes, large parts of Europe have become increasingly fragmented due to the expansion of urban and transport infrastructure (European Environment Agency (EEA), 2022 [↗](#)). This habitat loss and fragmentation can significantly impact water availability and trophic resources (Santos-Tovar et al., 2024 [↗](#)), which could in turn aid the spread and transmission of avian influenza viruses due to relocation of birds and outbreaks (Yin et al., 2022 [↗](#)). Climate change has also been reported as a factor that could affect wild bird distribution in different ways (Gilbert et al., 2008 [↗](#)). As global climate conditions change, avian migratory patterns and routes are also changing (Hitch and Leberg, 2007 [↗](#); Van Doren, 2022 [↗](#)). Furthermore, higher temperatures and extreme weather have resulted in large-scale population shifts in a range of temperate species (Dezfuli et al., 2022 [↗](#)). Overall, these elements therefore raise the question of the role of climate and land-use changes in the recent shifting epidemiology of HPAI in wild birds.

The increasing diversity of infected bird species is part of a broader resurgence of H5N1 globally. Phylogenetic analysis from Europe revealed significant genetic diversity in HPAI-H5N1 viruses since October 2020, likely due to multiple reassortment events with LPAI viruses. Notable introductions from Russia to Europe were observed through autumn migrations of wild birds (ECDC, 2021 [↗](#)). Additionally, the transatlantic transport of HPAI H5N1, documented in Canada in December 2021, suggests the virus can spread across continents, a route previously seen only with LPAI viruses (Caliendo et al., 2022 [↗](#); Dusek et al., 2014 [↗](#); Huang et al., 2014 [↗](#)). In November 2022, H5N1 (clade 2.3.4.4b) was introduced into South America, presumably through the movements of migratory wild birds travelling south during the boreal winter. The virus spread rapidly and caused significant mortality in wild birds and marine mammals across multiple countries (Leguia et al., 2023 [↗](#); Banyard et al., 2023 [↗](#); Campagna et al., 2024 [↗](#); Gamarra-Toledo et al., 2023 [↗](#); WOA, 2024 [↗](#)).

Interestingly, our models extrapolate environmental suitability for H5Nx in wild birds in areas where few or no outbreaks have been reported. This discrepancy may be explained by limited surveillance or underreporting in those regions. For instance, there is significant evidence that Kazakhstan and Central Asia play a role as a centre for the transmission of avian influenza viruses through migratory birds (Amirgazin et al., 2022 [↗](#); FAO, 2005 [↗](#); Sultankulova et al., 2024 [↗](#)). However, very few wild bird cases are reported in EMPRES-i. In contrast, Australia appears environmentally suitable in our models, yet no incursion of HPAI H5N1 2.3.4.4b has occurred

despite the arrival of millions of migratory shorebirds and seabirds from Asia and North America. Extensive surveillance in 2022 and 2023 found no active infections nor evidence of prior exposure to the 2.3.4.4b lineage (Wille et al., 2024 [DOI](#); Wille and Klaassen, 2023 [DOI](#)).

Despite the robustness of our ecological niche models, some limitations must be acknowledged. Firstly, the accuracy of our models is inherently dependent on the quality and completeness of the occurrence data used, which is influenced by the different reporting systems in place in different countries. Secondly, the definitions of key explanatory variables, such as intensive versus extensive production, may vary across regions, adding complexity to global predictions. To address these limitations, future work could focus on developing country-level or regional models that incorporate localised data and context-specific definitions. Thirdly, underreporting and biases in outbreak reports, particularly in wild bird populations, can skew model predictions and underestimate areas at risk. Furthermore, another limitation of our models for wild bird populations is the lack of incorporation of migratory patterns and behaviours. Migratory birds have complex movement patterns influenced by seasonal changes, food availability, and habitat conditions. Our models, using static spatial predictors, do not capture these dynamic routes and stopover sites, leading to potential inaccuracies in predicting areas at risk for HPAI circulation. In future work, the integration of dynamic migratory data and environmental changes into our ecological niche models could provide a more comprehensive and accurate risk assessment for HPAI spread among wild bird populations. In addition, aligning outbreak occurrences with seasonally matched environmental variables could further refine predictions of HPAI risk linked to migratory dynamics. Finally, our models for domestic poultry do not distinguish between primary introduction events (e.g., spillover from wild birds) and secondary transmission between farms due to limitations in the available surveillance data. While environmental factors likely influence the risk of initial spillover events, secondary spread is more often driven by anthropogenic factors such as biosecurity practices and poultry trade, which are not included in our current modelling framework.

Despite these limitations, our study provides significant contributions to the understanding of H5N1 and H5Nx HPAI virus dynamics. By integrating a wide range of spatial predictors and using ecological niche modelling techniques, we provide a quantitative analysis on the factors driving virus distribution. Overall, our study underscores the importance of continuous surveillance and the need for adaptive management strategies to address the evolving threat of HPAI viruses globally.

Materials and Methods

Our approach is similar to the one implemented by [Dhingra and colleagues \(2016\) DOI](#). While [Dhingra et al. \(2016\) DOI](#) developed their models only for domestic birds over the 2003-2016 periods, our models were developed for two host species separately (wild and domestic birds) and for two time periods (2016-2020 and 2020-2023).

HPAI occurrence data acquisition

The dataset containing highly pathogenic avian influenza (HPAI) occurrences, was constructed by extracting relevant information from the EMPRES-i database spanning the period from January 5, 2015, to March 7, 2023. This dataset encompasses comprehensive details pertaining to H5N1 and H5Nx subtypes, including the family of birds involved in transmission, observation and reporting dates, precise location coordinates, and the respective countries where the reports originated. The dataset comprises a total of 17,410 H5Nx and 8,383 H5N1 reported cases. It is important to note that the EMPRES-i database does not distinguish between index cases (e.g., primary spillover from wild birds) and secondary farm-to-farm transmissions. As such, our ecological niche models are

trained on confirmed HPAI outbreaks in poultry that may result from different transmission dynamics — including both initial introduction events influenced by environmental factors and subsequent spread within poultry systems.

Environmental data acquisition

Previous literature reviews (Gilbert and Pfeiffer, 2012; Dhingra et al., 2016) have summarised the predictor variables commonly linked to occurrences of HPAI. Here, we explored four sets of environmental variables considered by Dhingra and colleagues (Dhingra et al., 2016), and which are detailed in the table in Supplementary Information Resources S1: host variables (set 1), land cover variables (set 2), eco-climatic variables (set 3), and a risk-based selection of variables performed by Dhingra and colleagues (set 4).

Set 1 includes $\log_{10}$ -transformed extensive and intensive chicken density (Gilbert et al., 2015), domestic duck density (Robinson et al., 2014) and human population density from the Gridded Population of the World (Center For International Earth Science Information Network-CIESIN-Columbia University, 2017) database. The chicken density data includes GIS layers representing the global distribution of chickens, categorised into extensive and intensive systems. It uses the model published by Gilbert et al. (2018) applied to the Gridded Livestock of the World (GLW) version 3 chicken layers (Gilbert et al., 2022a). The global domestic duck distribution data were computed using the GLW version 4 (Gilbert et al., 2022b). Set 2 includes land cover information, which was obtained from the global 1-km Consensus Land Cover database by Tuanmu and Jetz, 2014, and identifies different land cover categories by providing a percentage of representation of each land cover category within raster cells of approximately one square kilometre (<http://www.earthenv.org/landcover.html>). Additionally, this dataset was supplemented by a layer about the distance of each spatial point to the open water. Set 3 encompasses seasonal and large-scale patterns of eco-climatic indices like daytime land surface temperature (day LST) and the normalised difference vegetation index (NDVI), as documented by Scharlemann and colleagues (Scharlemann et al., 2008). Finally, set 4 is a selection from the previous sets based on prior epidemiological knowledge (Gilbert and Pfeiffer, 2012; Dhingra et al., 2016).

Ecological niche modelling

The different ecological niche modelling analyses were performed with the boosted regression trees (BRT), available in the R package “dismo” (Hijmans et al., 2017). BRT is a machine learning approach that can be used to generate a collection of sequentially fitted regression trees optimising the predictive probability of occurrence given local environmental conditions (Elith et al., 2006). The predictive probability, interpreted as ecological suitability for local virus circulation, ranges from “0” (unsuitable environmental conditions) and “1” (highly suitable environmental conditions). BRT allows us to model complex non-linear relationships between the response and various predictor variables (Elith et al., 2008). It has also been demonstrated that the BRT approach had a superior predictive performance compared to alternative modelling approaches (Elith et al., 2006).

We selected 2020 as a cut-off point to reflect a well-documented shift in HPAI epidemiology, notably the emergence and global spread of clade 2.3.4.4b. This event marked a turning point in viral dynamics, influencing both the range of susceptible hosts and the geographical distribution of outbreaks. Based on this distinction, we performed ten replicated BRT analyses using each of the seven occurrence datasets: (i) H5Nx cases in wild birds < 2020 (01/01/2015 - 31/12/2019), (ii) H5N1 cases in wild birds > 2020 (01/01/2020 - 06/03/2023), (iii) H5Nx cases in wild birds > 2020 (01/01/2020 - 06/03/2023), (iv) H5Nx cases in domestic birds < 2020 (01/01/2015 - 31/12/2019), (v) H5N1 cases in domestic birds < 2020 (01/01/2015 - 31/12/2019), (vi) H5N1 cases in domestic birds > 2020 (01/01/2020 - 06/03/2023), and (vii) H5Nx cases in domestic birds > 2020 (01/01/2020 - 06/03/2023). Of note, there was not enough occurrence data to train ecological niche models on

H5N1 cases in wild birds < 2020 (only 33 occurrence data from eight different countries — Bangladesh, Bulgaria, China, India, Kazakhstan, Nepal, Romania, and Russia — between the 01/01/2015 and 31/12/2019).

We ran BRT analyses using both occurrence and “pseudo-absence” data, the latter corresponding to randomly sampled absence points within the study area (Elith et al., 2006 [↗](#)). To account for heterogeneity in AIV surveillance and minimise the risk of sampling pseudo-absences in poorly monitored regions, we restricted our analysis to countries (or administrative level 1 units in China and Russia) with at least five confirmed outbreaks. Unlike Dhingra et al. (2016) [↗](#), who sampled pseudo-absences across a broader global extent, our sampling was limited to regions with demonstrated surveillance activity. In addition, we adjusted the density of pseudo-absence points according to the number of reported outbreaks in each country or admin-1 unit, as a proxy for surveillance effort — an approach not implemented in this previous study.

In addition, the sampling pseudo-absence points was based on the human population distribution to further account for potential within country differences in surveillance intensity (Dhingra et al., 2016 [↗](#)): the probability to sample a pseudo-absence from a given grid cell was proportional to the log-transformed human population density in that cell. The human population density raster was retrieved from the Gridded Population of the World (GPW) project, which was downscaled to a resolution of 5 arcminutes. Another approach to sampling pseudo-absences would have been to distribute them according to the density of domestic poultry. However, as pointed out by Dhingra and colleagues (2016) [↗](#), the locations of outbreaks in the EMPRES-i database are often georeferenced using place name nomenclatures due to a lack of accurate GPS data, which could introduce a spatial bias towards populated areas. Therefore, the use of human population density as an indicator of surveillance effort takes into account not only the probability of detection, but also the geo-referencing bias in household records. To further reduce bias from unsuitable regions where the pathogen may not have been introduced, we restricted pseudo-absence sampling to areas located at a minimum of 10 km and a maximum of 1,000 km from known presence points (Phillips et al., 2009; Dhingra et al., 2016 [↗](#)). Initially, ten times more pseudo-absences than presence points were simulated, but we ultimately retained three times more pseudo-absences than presences in each country/admin-1 unit.

Given that spatial data typically exhibits autocorrelation, employing a standard cross-validation procedure may result in an overestimation of the BRT model’s accuracy (Randin et al., 2006). To circumvent this issue, we employed a spatial cross-validation procedure to select the optimal number of trees in each BRT model. Specifically, we tested and compared two distinct spatial cross-validation procedures: (i) the procedure introduced by Dhingra and colleagues and based on (five) reference presence points used to cluster the presence and pseudo-absence points in (five) folds according to their nearest distance to a reference point, and (ii) the procedure introduced by Valavi and colleagues and based on a block generation method (Valavi et al., 2019 [↗](#)). In the first spatial cross-validation procedure, the five reference presence points were randomly selected among all presence points. If each of the five generated spatial folds did not gather at least 10% of the presence points, the random selection of reference points was re-initiated. To compare the performance of the three cross-validation procedures applied in this study (the standard procedure and the spatial procedures based on the reference points or blocks generation), we estimated the spatial sorting bias (SSB) metric (Hijmans, 2012 [↗](#)) ranging from 0 to 1, an SSB near 0 and 1 indicating an extreme impact and the absence of an impact of spatial autocorrelation on the model training, respectively.

All BRT analyses were run and averaged over 10 cross-validated replicates, with a tree complexity of 4, a learning rate of 0.01, a tolerance parameter of 0.001, and while considering 5 spatial folds. Each model was initiated with 10 trees, and additional trees were incrementally added (in steps of 5) up to a maximum of 10,000, with the optimal number selected based on cross-validation tests. We evaluated the inferences using the area under the receiver operating characteristic (ROC)

curve, also simply referred to as “area under the curve” (AUC). We further used the AUC metric to assess the capacity of our models trained on occurrence data < 2020 to predict > 2020 distribution of occurrence data. To assess how each environmental factor contributed to the different BRT models, we computed their relative influence (RI) in those models and their response curves showing how ecological suitability varies with one specific factor. RI values were obtained by evaluating the number of times a specific environmental factor was selected for splitting a tree, weighted by the squared improvement to the model as a result of each split, averaged over all trees (Elith et al., 2008).

We explored the four sets of environmental variables assembled by Dhingra and colleagues (Dhingra et al., 2016 [↗](#)) and described above in the “Environmental data acquisition” subsection. The BRT analyses presented here were eventually performed based on set 2 for wild bird cases and set 4 for domestic bird cases. This selection is aligned with the methodology of Dhingra and colleagues (2016) [↗](#), prioritising variables which demonstrated the best predictive accuracy and clear epidemiological significance to minimise the potential for coincidental correlations. The RI values obtained when analysing each set of environmental variables separately are all reported in a table in Supplementary Information Resources S2.

Bird diversity indices calculation

Diversity index calculations were performed using the HPAI dataset. Cases lacking information on the date of collection, bird family, and subtype were excluded from the analyses. The diversity of HPAI cases across bird families was quantified using two indices: the Shannon diversity index and the Simpson diversity index. The Shannon index accounts for evenness of bird families (Shannon, 1948 [↗](#)), whereas the Simpson index provides a measure of dominance (Simpson, 1949 [↗](#)), emphasising the most abundant families in the dataset. Simpson index ranges between 0 and 1, where 1 represents infinite diversity and 0 no diversity. Calculations were done using two time points: before and after 2020, and for various data classifications: overall and annually, across all bird families and specifically sea bird families, as well as for all subtypes (HxNx) and those identified as H5N1 subtype, wild birds, and poultry. These indices were calculated using the R package “vegan” (Oksanen et al., 2023): the dataset was first aggregated by year, HPAI subtype and bird family, and we then used the “diversity” function to compute the Shannon and Simpson indices for each year.

Data and code accessibility

R scripts and related files needed to run the analyses, as well as Supplementary Information Resources S1 (description and source of each environmental variable included in the original sets of variables) and S2 (mean relative influence computed for the environmental variables within each considered set of variables), are all available at https://github.com/sdellicour/h5nx_risk_mapping [↗](#).

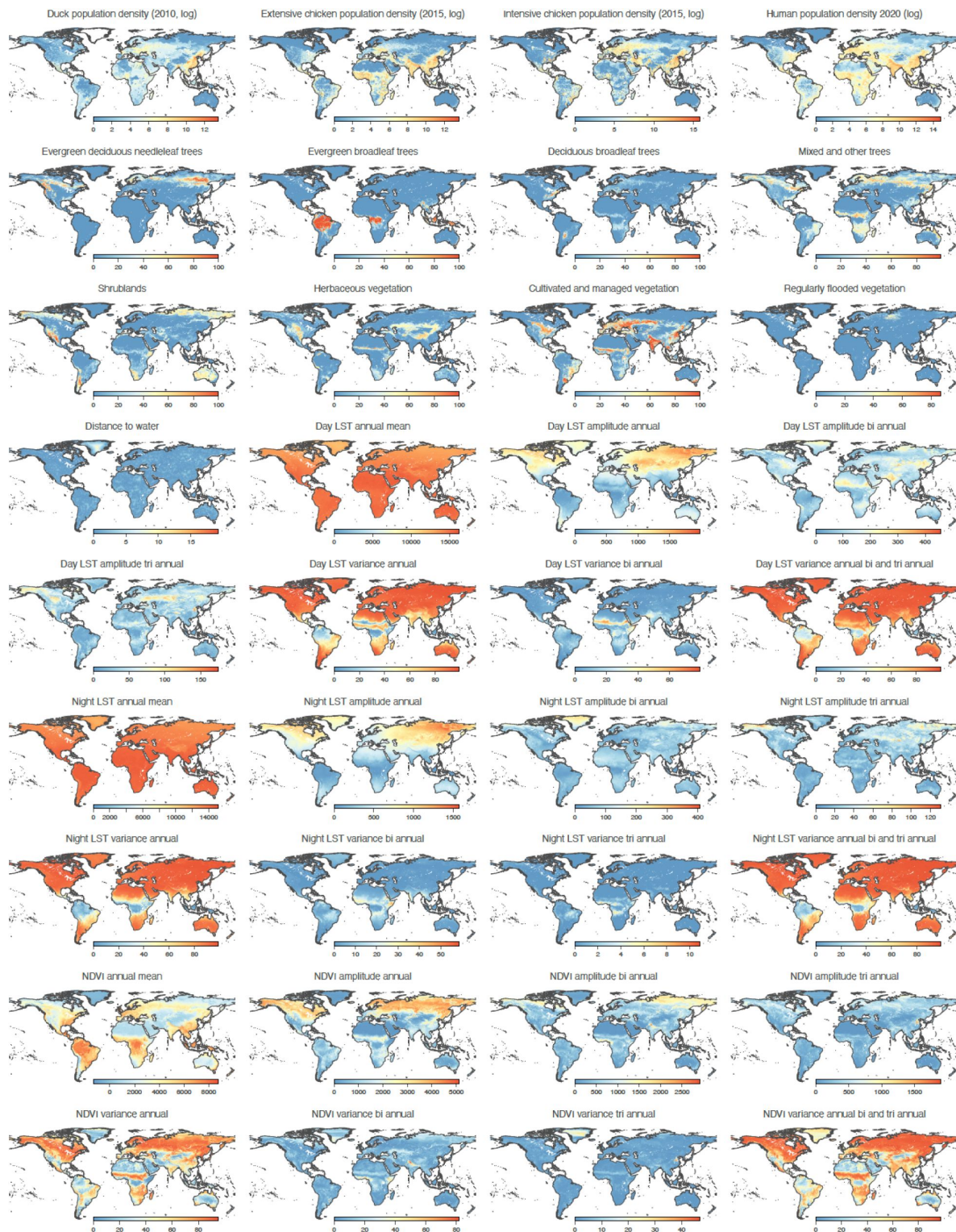


Figure S1

Environmental factors included in the ecological niche modelling.

“LST” refers to “land surface temperature”, “NDVI” to the “normalised difference vegetation index”, and “EVI” to the “enhanced vegetation index”.

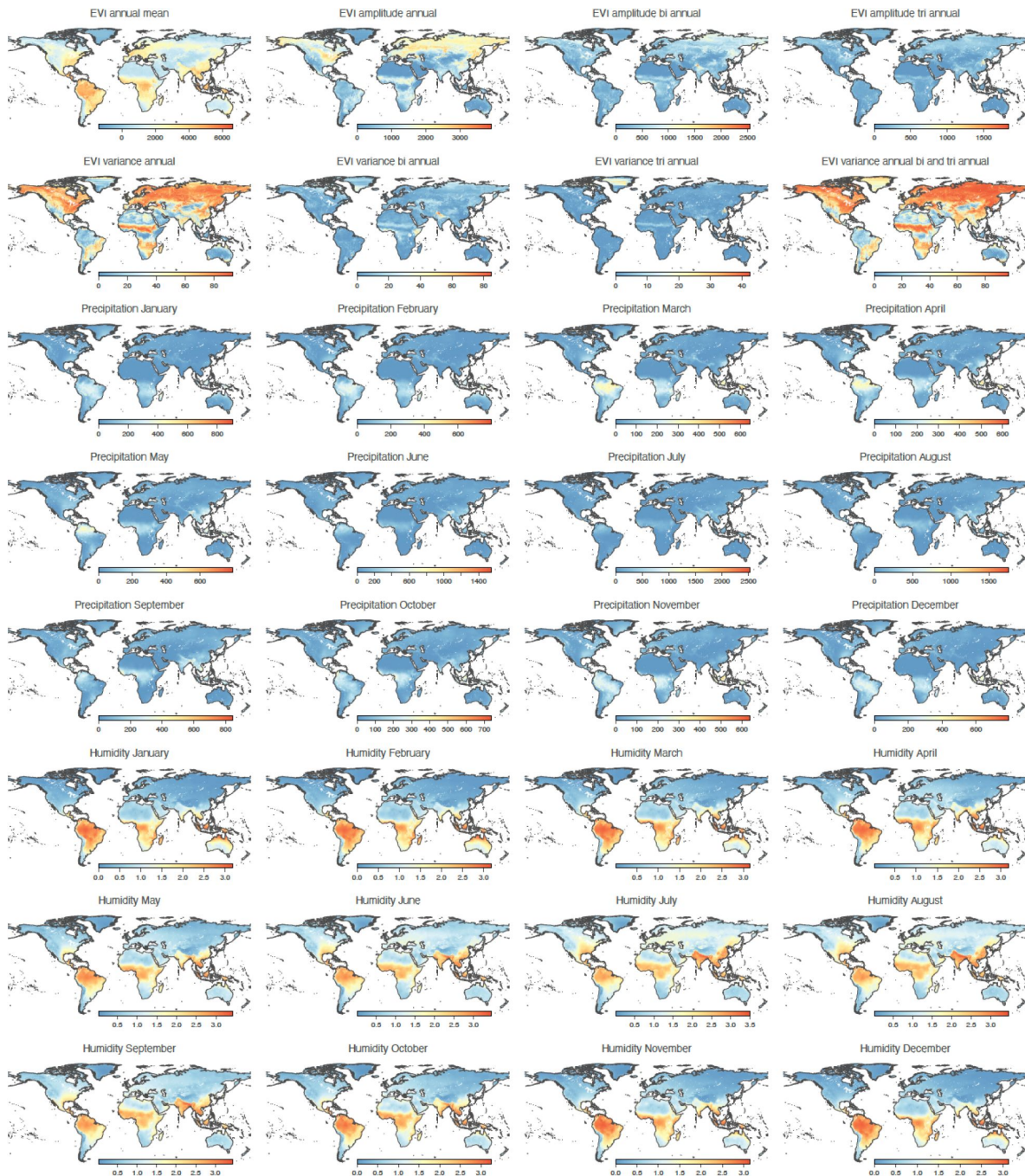


Figure S1 (continued)

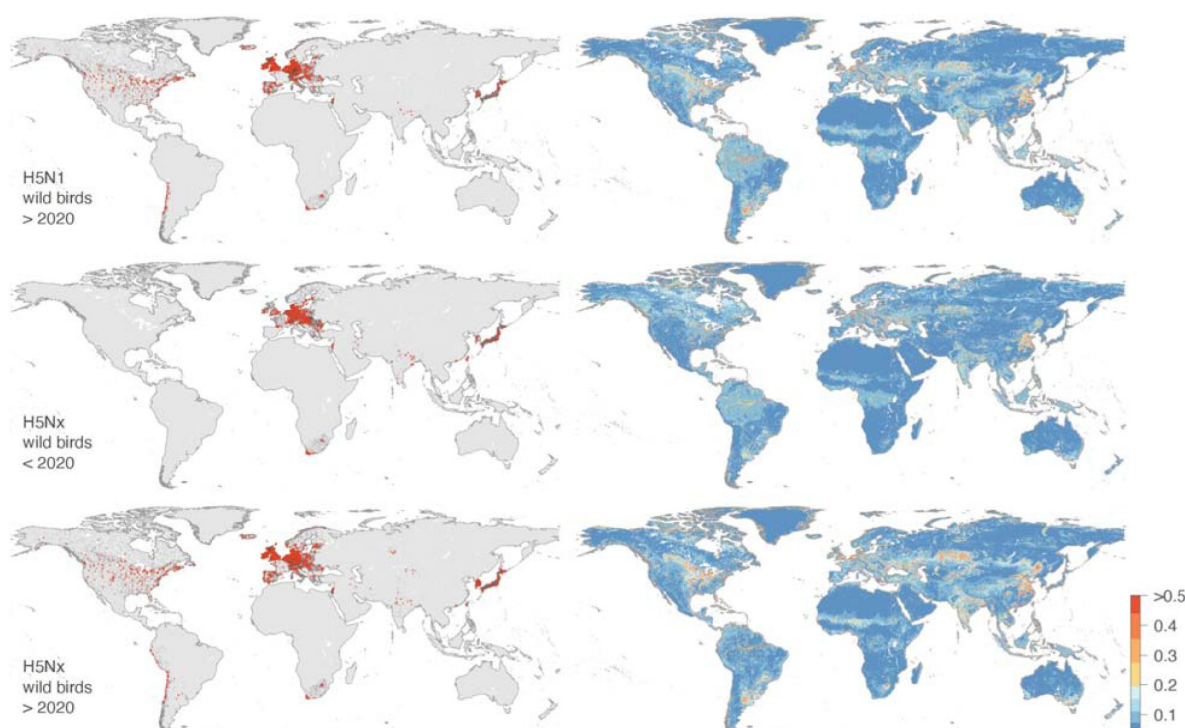


Figure S2

Comparison between the presence/pseudo-absence data used to train the ecological niche models (1° column) and the resulting ecological niche suitability maps (2° column).

In the first column, presence and sampled pseudo-absence points are displayed as red and dark points, respectively. As detailed in the Materials and Methods section, pseudo-absence points were only sampled in countries in which there was at least one presence point (for either H5N1 or H5Nx, the considering time period, and bird populations). For Russia and China, we considered the admin-1 administrative areas instead of the country to define if pseudo-absence points should be sampled in the corresponding area. Specifically, for a given administrative region (i.e. a given country or admin-1 region in case of Russia or China) gathering at least one presence point, we sampled a number of pseudo-absence points equal to three times the number of presence points in that region. Furthermore, pseudo-absence points were sampled according to the log-transformed raster of human population density, which prevented the sampling of pseudo-absence in unpopulated areas non associated with any surveillance activity.

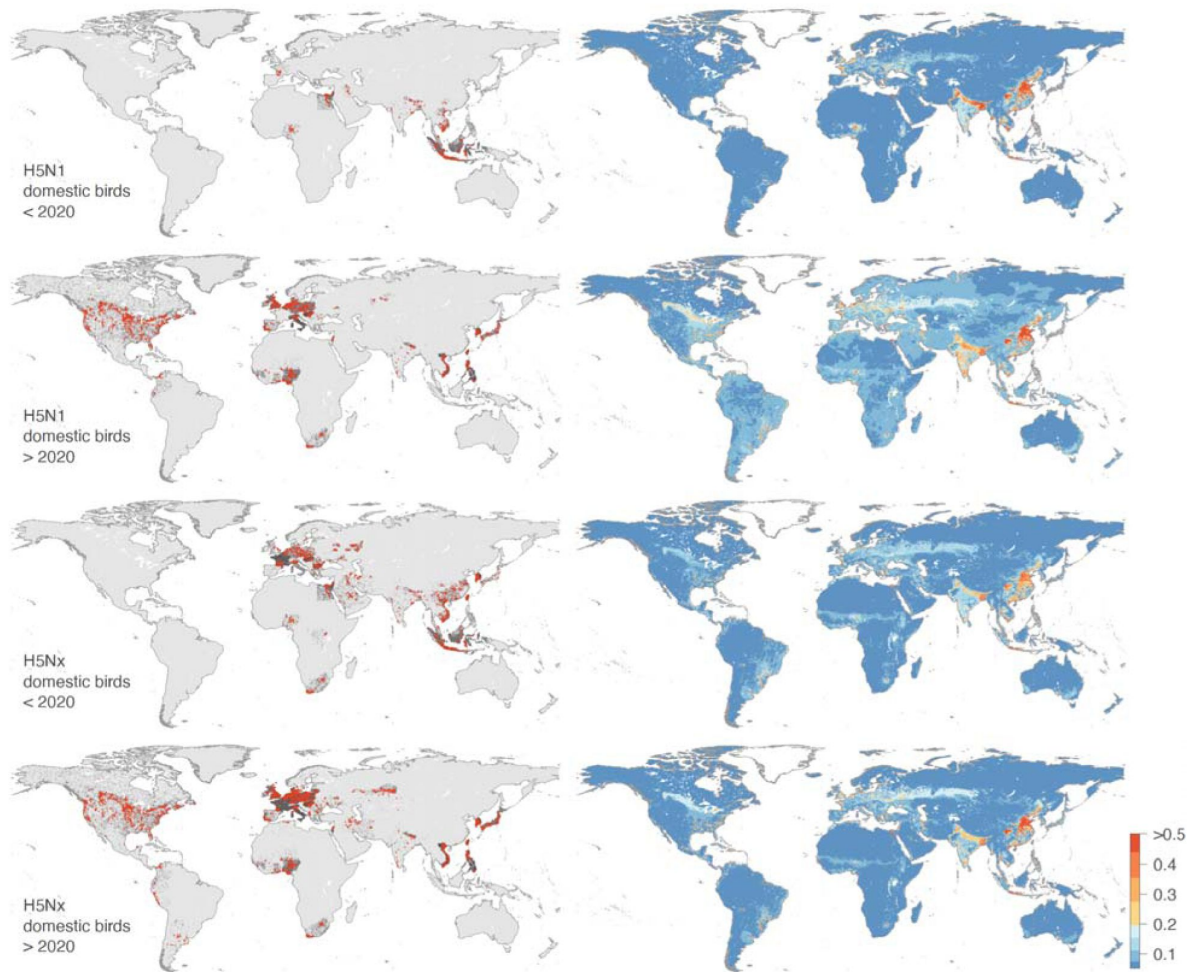


Figure S2 (continued)

Figure S3

Boxplots reporting the sorting sampling bias (SSB) and area under the curve (AUC) of the receiving operator curve metrics associated with various ecological niche models trained for H5N1 and H5Nx.

For the SSB estimates, we report values obtained for the standard cross-validation (A), as well as spatial cross-validation procedures based on the reference points (B) and the blocks generation (C) approaches (see the Materials and Methods section for further detail). SSB estimates range from 0 to 1; values close to 0 indicate a strong impact of spatial autocorrelation on model training, while values close to 1 suggest little to no impact. The AUC metrics computed for the ecological niche models trained with the spatial cross-validation procedures based on the “epicentres”, and we here report the values obtained for the models trained on each of the four distinct sets of environmental factors.

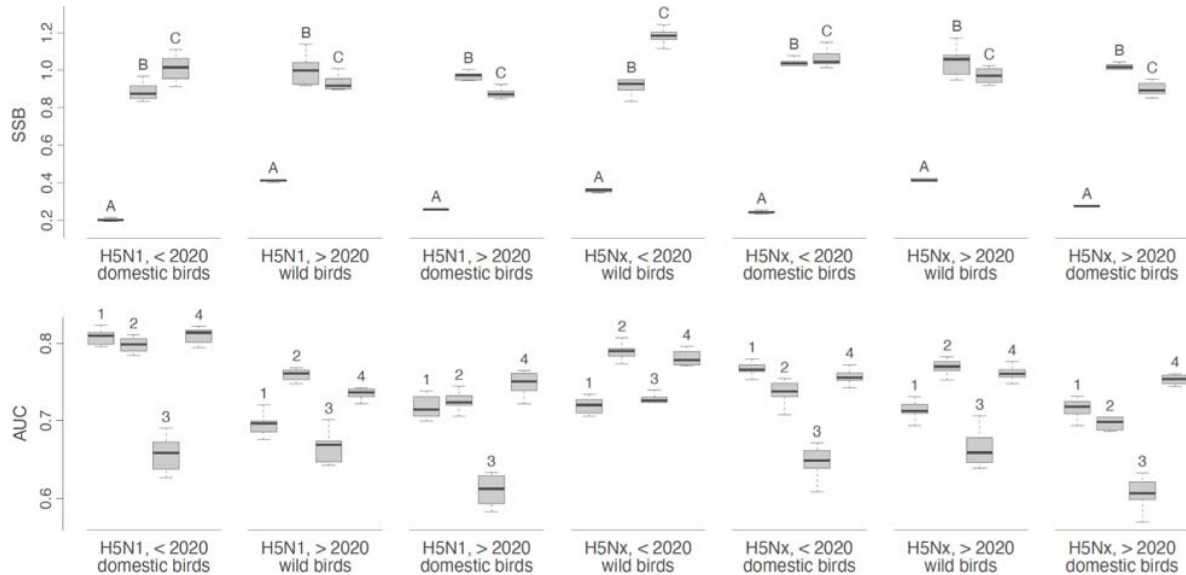


Figure S4

Distribution of H5Nx and H5N1 occurrence records in sea birds from 2015 to 2023, categorised by bird family.

The three panels successively show the total occurrence records for all H5Nx subtypes, occurrence records for non-H5N1 H5Nx subtypes, and all H5N1 occurrence records.

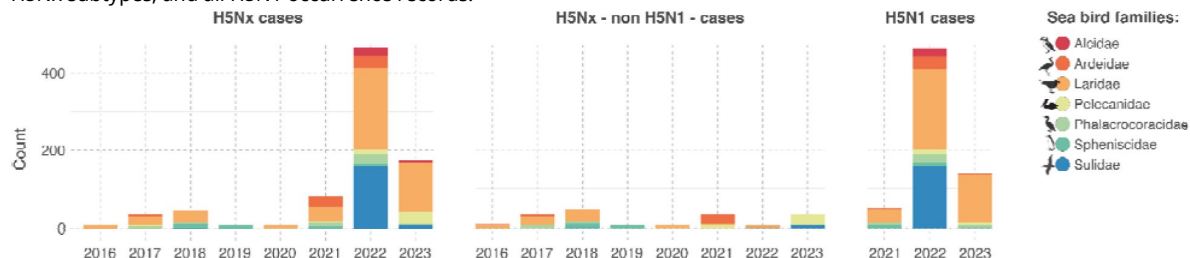


Figure S5

Comparison between the ecological niche suitability estimated for H5N1 and H5Nx before 2016 and after 2020 for domestic bird populations.

Specifically, we here compare the ecological niche models trained by [Dhingra et al. \(2016\)](#) on occurrence records collected between January 2004 to March 2015 (< 2016) and the ecological niche models trained in the present study on occurrence records collected between January 2020 to March 2023 (> 2020).

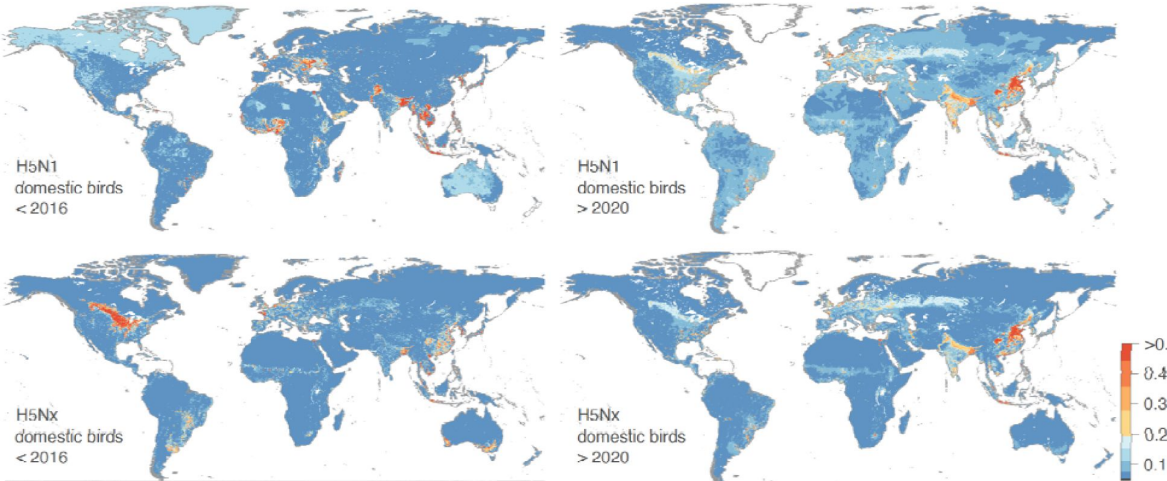


Table S1

Assessment of the predictive performance of ecological niche models to predict present and past ecological suitability maps.

We here report, for each replicate analysis, a measure of the predictive performance of the trained ecological niche through the computation of the area under the receiver operating characteristic (ROC) curve, also simply referred to as “area under the curve” (AUC). (*) refers to AUC estimates obtained when assessing the capacity of models trained on occurrence data < 2020 to predict > 2020 distribution of occurrence data.

	Wild birds			Domestic birds					
	H5Nx			H5N1			H5Nx		
	< 2020	> 2020	>2020*	< 2020	> 2020	> 2020*	< 2020	> 2020	> 2020*
Replicate 1	0.899	0.793	0.700	0.832	0.812	0.769	0.833	0.791	0.755
Replicate 2	0.896	0.769	0.703	0.832	0.822	0.769	0.829	0.781	0.752
Replicate 3	0.895	0.796	0.707	0.834	0.820	0.770	0.818	0.784	0.743
Replicate 4	0.894	0.773	0.705	0.843	0.820	0.769	0.828	0.804	0.751
Replicate 5	0.893	0.797	0.706	0.837	0.816	0.770	0.828	0.780	0.752
Replicate 6	0.898	0.797	0.701	0.845	0.816	0.770	0.829	0.780	0.752
Replicate 7	0.897	0.798	0.700	0.835	0.816	0.769	0.827	0.775	0.750
Replicate 8	0.894	0.805	0.705	0.836	0.825	0.770	0.828	0.777	0.752
Replicate 9	0.894	0.770	0.702	0.834	0.813	0.769	0.828	0.779	0.750
Replicate 10	0.894	0.775	0.703	0.835	0.823	0.769	0.832	0.795	0.754

	Wild birds			
	H5N1		H5Nx	
	< 2020	> 2020	< 2020	> 2020
Evergreen deciduous needleleaf trees	-	1.6 % [1.5-1.6]	1.6 % [1.3-1.7]	0.4 % [0.4-0.5]
Evergreen broadleaf trees	-	1.3 % [1.2-1.4]	0.1 % [0.1-0.1]	1.0 % [0.8-1.1]
Deciduous broadleaf trees	-	4.1 % [4.0-4.2]	1.6 % [1.5-1.8]	6.1 % [5.9-6.2]
Mixed and other trees	-	11.5 % [11.4-11.7]	5.3 % [4.9-5.4]	11.4 % [11.3-11.8]
Shrublands	-	2.3 % [2.0-2.3]	3.2 % [3.0-3.3]	2.4 % [2.1-2.7]
Herbaceous vegetation	-	8.8 % [8.8-9.0]	2.8 % [2.5-3.0]	9.5 % [9.5-9.6]
Cultivated and managed vegetation	-	4.4 % [4.3-4.5]	4.8 % [4.6-4.9]	5.8 % [5.4-6.1]
Regularly flooded vegetation	-	0.3 % [0.3-0.3]	0.1 % [0.1-0.1]	0.2 % [0.1-0.3]
Urban and built-up areas	-	39.3 % [39.0-39.9]	54.5 % [53.5-55.0]	37.5 % [36.5-38.2]
Open water areas	-	22.0 % [21.9-22.3]	25.5 % [25.3-25.8]	23 % [22.8-23.3]
Distance to water	-	4.1 % [3.9-4.3]	0.9 % [0.7-1.0]	2.8 % [2.4-3.1]
	Domestic birds			
	H5N1		H5Nx	
	< 2020	> 2020	< 2020	> 2020
Duck pop. density (2010, log)	18.2 % [17.8-18.4]	15.7 % [13.3-16.4]	28.7 % [28.4-28.8]	12.9 % [12.9-13.5]
Extensive chicken pop. density (2015, log)	4.4 % [4.2-4.5]	2.3 % [0.5-2.6]	3.9 % [3.7-4.0]	2.7 % [2.5-3.4]
Intensive chicken pop. density (2015, log)	8.5 % [8.4-8.7]	30.4 % [29.6-36.2]	29.6 % [29.3-30.1]	43.1 % [40.5-43.4]
Human pop. density (2020, log)	54.9 % [53.9-55.1]	20.7 % [20.4-21.6]	15.5 % [15.3-15.7]	14.9 % [14.8-15.2]
Cultivated and managed vegetation	9.5 % [9.3-9.7]	16.6 % [16.3-18.5]	7.1 % [7.0-7.2]	13.2 % [12.9-13.3]
Open water areas	0.5 % [0.5-0.6]	1.4 % [0.3-1.5]	0.0 % [0.0-0.0]	0.6 % [0.4-0.8]
Day LST annual mean	4.1 % [4.0-4.2]	12.9 % [9.6-13.3]	15.4 % [15.1-15.6]	13.0 % [12.4-13.9]

Table S2

Relative influence (RI, in %) of each environmental variable in the respective ecological models trained on a specific dataset of occurrence data.

A RI estimate was obtained for each replicate BRT analysis and we here report median as well as first and third quartile RI values.

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Reviewer #1 (Public review):

Summary:

The authors aim to predict ecological suitability for transmission of highly pathogenic avian influenza (HPAI) using ecological niche models. This class of models identify correlations between the locations of species or disease detections and the environment. These correlations are then used to predict habitat suitability (in this work, ecological suitability for disease transmission) in locations where surveillance of the species or disease has not been conducted. The authors fit separate models for HPAI detections in wild birds and farmed birds, for two strains of HPAI (H5N1 and H5Nx) and for two time periods, pre- and post-2020. The authors also validate models fitted to disease occurrence data from pre-2020 using post-2020 occurrence data. I thank the authors for taking the time to respond to my initial review and I provide some follow-up below.

Detailed comments:

In my review, I asked the authors to clarify the meaning of "spillover" within the HPAI transmission cycle. This term is still not entirely clear: at lines 409-410, the authors use the term with reference to transmission between wild birds and farmed birds, as distinct to transmission between farmed birds. It is implied but not explicitly stated that "spillover" is relevant to the transmission cycle in farmed birds only. The sentence, "we developed separate ecological niche models for wild and domestic bird HPAI occurrences ..." could have been supported by a clear sentence describing the transmission cycle, to prime the reader for why two separate models were necessary.

I also queried the importance of (dead-end) mammalian infections to a model of the HPAI transmission risk, to which the authors responded: "While spillover events of HPAI into mammals have been documented, these detections are generally considered dead-end infections and do not currently represent sustained transmission chains. As such, they fall outside the scope of our study, which focuses on avian hosts and models ecological suitability for outbreaks in wild and domestic birds." I would argue that any infections, whether they are in dead-end or competent hosts, represent the presence of environmental conditions to support transmission so are certainly relevant to a niche model and therefore within scope. It is certainly understandable if the authors have not been able to access data of mammalian infections, but it is an oversight to dismiss these infections as irrelevant.

Correlative ecological niche models, including BRTs, learn relationships between occurrence data and covariate data to make predictions, irrespective of correlations between covariates. I am not convinced that the authors can make any "interpretation" (line 298) that the covariates that are most informative to their models have any "influence" (line 282) on their response variable. Indeed, the observation that "land-use and climatic predictors do not play an important role in the niche ecological models" (line 286), while "intensive chicken population density emerges as a significant predictor" (line 282) begs the question: from an

operational perspective, is the best (e.g., most interpretable and quickest to generate) model of HPAI risk a map of poultry farming intensity?

I have more significant concerns about the authors' treatment of sampling bias: "We agree with the Reviewer's comment that poultry density could have potentially been considered to guide the sampling effort of the pseudo-absences to consider when training domestic bird models. We however prefer to keep using a human population density layer as a proxy for surveillance bias to define the relative probability to sample pseudo-absence points in the different pixels of the background area considered when training our ecological niche models. Indeed, given that poultry density is precisely one of the predictors that we aim to test, considering this environmental layer for defining the relative probability to sample pseudo-absences would introduce a certain level of circularity in our analytical procedure, e.g. by artificially increasing to influence of that particular variable in our models." The authors have elected to ignore a fundamental feature of distribution modelling with occurrence-only data: if we include a source of sampling bias as a covariate and do not include it when we sample background data, then that covariate would appear to be correlated with presence. They acknowledge this later in their response to my review: "...assuming a sampling bias correlated with poultry density would result in reducing its effect as a risk factor." In other words, the apparent predictive capacity of poultry density is a function of how the authors have constructed the sampling bias for their models. A reader of the manuscript can reasonably ask the question: to what degree is the model a model of HPAI transmission risk, and to what degree is the model a model of the observation process? The sentence at lines 474-477 is a helpful addition, however the preceding sentence, "Another approach to sampling pseudo-absences would have been to distribute them according to the density of domestic poultry," (line 474) is included without acknowledgement of the flow-on consequence to one of the key findings of the manuscript, that "...intensive chicken population density emerges as a significant predictor..." (line 282). The additional context on the EMPRES-i dataset at line 475-476 ("the locations of outbreaks ... are often georeferenced using place name nomenclatures") is in conflict with the description of the dataset at line 407 ("precise location coordinates"). Ultimately, the choices that the authors have made are entirely defensible through a clear, concise description of model features and assumptions, and precise language to guide the reader through interpretation of results. I am not satisfied that this is provided in the revised manuscript.

The authors have slightly misunderstood my comment on "extrapolation": I referred to "environmental extrapolation" in my review without being particularly explicit about my meaning. By "environmental extrapolation", I meant to ask whether the models were predicting to environments that are outside the extent of environments included in the occurrence data used in the manuscript. The authors appear to have understood this to be a comment on geographic extrapolation, or predicting to areas outside the geographic extent included in occurrence data, e.g.: "For H5Nx post-2020, areas of high predicted ecological suitability, such as Brazil, Bolivia, the Caribbean islands, and Jilin province in China, likely result from extrapolations, as these regions reported few or no outbreaks in the training data" (lines 195-197). Is the model extrapolating in environmental space in these regions? This is unclear. I do not suggest that the authors should carry out further analysis, but the multivariate environmental similarity surface (MESS; see Elith et al., 2010: <https://doi.org/10.1111/j.2041-210X.2010.00036.x>) is a useful tool to visualise environmental extrapolation and aid model interpretation.

On the subject of "extrapolation", I am also concerned by the additions at lines 362-370: "...our models extrapolate environmental suitability for H5Nx in wild birds in areas where few or no outbreaks have been reported. This discrepancy may be explained by limited surveillance or underreporting in those regions." The "discrepancy" cited here is a feature of the input dataset, a function of the observation distribution that should be captured in pseudo-absence data. The authors state that Kazakhstan and Central Asia are areas of interest, and that the

environments in this region are outside the extent of environments captured in the occurrence dataset, although it is unclear whether "extrapolation" is informed by a quantitative tool like a MESS or judged by some other qualitative test. The authors then cite Australia as an example of a region with some predicted suitability but no HPAI outbreaks to date, however this discussion point is not linked to the idea that the presence of environmental conditions to support transmission need not imply the occurrence of transmission (as in the addition, "...spatial isolation may imply a lower risk of actual occurrences..." at line 214). Ultimately, the authors have not added any clear comment on model uncertainty (e.g., variation between replicated BRTs) as I suggested might be helpful to support their description of model predictions.

All of my criticisms are, of course, applied with the understanding that niche modelling is imperfect for a disease like HPAI, and that data may be biased/incomplete, etc.: these caveats are common across the niche modelling literature. However, if language around the transmission cycle, the niche, and the interpretation of any of the models is imprecise, which I find it to be in the revised manuscript, it undermines all of the science that is presented in this work.

<https://doi.org/10.7554/eLife.104748.2.sa2>

Reviewer #2 (Public review):

Summary:

The geographic range of highly pathogenic avian influenza cases changed substantially around the period 2020, and there is much interest in understanding why. Since 2020 the pathogen irrupted in the Americas and the distribution in Asia changed dramatically. This study aimed to determine which spatial factors (environmental, agronomic and socio-economic) explain the change in numbers and locations of cases reported since 2020 (2020–2023). That's a causal question which they address by applying correlative environmental niche modelling (ENM) approach to the avian influenza case data before (2015–2020) and after 2020 (2020–2023) and separately for confirmed cases in wild and domestic birds. To address their questions they compare the outputs of the respective models, and those of the first global model of the HPAI niche published by Dhingra et al 2016.

ENM is a correlative approach useful for extrapolating understandings based on sparse geographically referenced observational data over un- or under-sampled areas with similar environmental characteristics in the form of a continuous map. In this case, because the selected covariates about land cover, use, population and environment are broadly available over the entire world, modelled associations between the response and those covariates can be projected (predicted) back to space in the form of a continuous map of the HPAI niche for the entire world.

Strengths:

The authors are clear about expected bias in the detection of cases, such geographic variation in surveillance effort (testing of symptomatic or dead wildlife, testing domestic flocks) and in general more detections near areas of higher human population density (because if a tree falls in a forest and there is no-one there, etc), and take steps to ameliorate those. The authors use boosted regression trees to implement the ENM, which typically feature among the best performing models for this application (also known as habitat suitability models). They ran replicate sets of the analysis for each of their model targets (wild/domestic x pathogen variant), which can help produce stable predictions. Their code and data is provided, though I did not verify that the work was reproducible.

The paper can be read as a partial update to the first global model of H5Nx transmission by Dhingra and others published in 2016 and explicitly follows many methodological elements. Because they use the same covariate sets as used by Dhingra et al 2016 (including the comparisons of the performance of the sets in spatial cross-validation) and for both time periods of interest in the current work, comparison of model outputs is possible. The authors further facilitate those comparisons with clear graphics and supplementary analyses and presentation. The models can also be explored interactively at a weblink provided in text, though it would be good to see the model training data there too.

The authors' comparison of ENM model outputs generated from the distinct HPAI case datasets is interesting and worthwhile, though for me, only as a response to differently framed research questions.

Weaknesses:

This well-presented and technically well-executed paper has one major weakness to my mind. I don't believe that ENM models were an appropriate tool to address their stated goal, which was to identify the factors that "explain" changing HPAI epidemiology.

Here is how I understand and unpack that weakness:

(1) Because of their fundamentally correlative nature, ENMs are not a strong candidate for exploring or inferring causal relationships.

(2) Generating ENMs for a species whose distribution is undergoing broad scale range change is complicated and requires particular caution and nuance in interpretation (e.g., Elith et al, 2010, an important general assumption of environmental niche models is that the target species is at some kind of distributional equilibrium (at time scales relevant to the model application). In practice that means the species has had an opportunity to reach all suitable habitats and therefore its absence from some can be interpreted as either unfavourable environment or interactions with other species). Here data sets for the response (N5H1 or N5Hx case data in domestic or wild birds) were divided into two periods; 2015–2020, and 2020–2023 based on the rationale that the geographic locations and host-species profile of cases detected in the latter period was suggestive of changed epidemiology. In comparing outputs from multiple ENMs for the same target from distinct time periods the authors are expertly working in, or even dancing around, what is a known grey area, and they need to make the necessary assumptions and caveats obvious to readers.

(3) To generate global prediction maps via ENM, only variables that exist at appropriate resolution over the desired area can be supplied as covariates. What processes could influence changing epidemiology of a pathogen and are their covariates that represent them? Introduction to a new geographic area (continent) with naive population, immunity in previously exposed populations, control measures to limit spread such as vaccination or destruction of vulnerable populations or flocks? Might those control measures be more or less likely depending on the country as a function of its resources and governance? There aren't globally available datasets that speak to those factors, so the question is not why were they omitted but rather was the authors decision to choose ENMs given their question justified? How valuable are insights based on patterns of correlation change when considering different temporal sets of HPAI cases in relation to a common and somewhat anachronistic set of covariates?

(4) In general the study is somewhat incoherent with respect to time. Though the case data come from different time periods, each response dataset was modelled separately using exactly the same covariate dataset that predated both sets. That decision should be understood as a strong assumption on the part of the authors that conditions the interpretation: the world (as represented by the covariate set) is immutable, so the model has

to return different correlative associations between the case data and the covariates to explain the new data. While the world represented by the selected covariates *may* be relatively stable (could be statistically confirmed), what about the world not represented by the covariates (see point 3)?

References:

Dhingra et al, 2016, Global mapping of highly pathogenic avian influenza H5N1 and H5Nx clade 2.3.4.4 viruses with spatial cross-validation, eLife 5, <https://doi.org/10.7554/eLife.19571>

Elith, J., Kearney, M., & Phillips, S. (2010). The art of modelling range-shifting species. *Methods in Ecology and Evolution*, 1(4), 330-342.

<https://doi.org/10.7554/eLife.104748.2.sa1>

Author response:

The following is the authors' response to the current reviews.

Public Reviews:

We thank the Reviewers for their thorough attention to our paper and the interesting discussion about the findings. Before responding to more specific comments, here some general points we would like to clarify:

(1) Ecological niche models are indeed correlative models, and we used them to highlight environmental factors associated with HPAI outbreaks within two host groups. We will further revise the terminology that could still unintentionally suggest causal inference. The few remaining ambiguities were mainly in the Discussion section, where our intent was to interpret the results in light of the broader scientific literature. Particularly, we will change the following expressions:

- “Which factors can explain...” to “Which factors are associated with...” (line 75);
- “the environmental and anthropogenic factors influencing” to “the environmental and anthropogenic factors that are correlated with” (line 273);
- “underscoring the influence” to “underscoring the strong association” (line 282).

(2) We respectfully disagree with the suggestion that an ecological niche modelling (ENM) approach is not appropriate for this work and the research question addressed therein. Ecological niche models are specifically designed to estimate the spatial distribution of the environmental suitability of species and pathogens, making them well suited to our research questions. In our study, we have also explicitly detailed the known limitations of ecological niche models in the Discussion section, in line with prior literature, to ensure their appropriate interpretation in the context of HPAI.

(3) The environmental layers used in our models were restricted to those available at a global scale, as listed in Supplementary Information Resources S1 (https://github.com/sdellicour/h5nx_risk_mapping/blob/master/Scripts_%26_data/SI_Resource_S1.xlsx). Naturally, not all potentially relevant environmental factors could be included, but the selected layers are explicitly documented and only these were assessed for their importance. Despite this limitation, the performance metrics indicate that the models performed well, suggesting that the chosen covariates capture meaningful associations with HPAI occurrence at a global scale.

Reviewer #1 (Public review):

The authors aim to predict ecological suitability for transmission of highly pathogenic avian influenza (HPAI) using ecological niche models. This class of models identify correlations between the locations of species or disease detections and the environment. These correlations are then used to predict habitat suitability (in this work, ecological suitability for disease transmission) in locations where surveillance of the species or disease has not been conducted. The authors fit separate models for HPAI detections in wild birds and farmed birds, for two strains of HPAI (H5N1 and H5Nx) and for two time periods, pre- and post-2020. The authors also validate models fitted to disease occurrence data from pre-2020 using post-2020 occurrence data. I thank the authors for taking the time to respond to my initial review and I provide some follow-up below.

Detailed comments:

In my review, I asked the authors to clarify the meaning of "spillover" within the HPAI transmission cycle. This term is still not entirely clear: at lines 409-410, the authors use the term with reference to transmission between wild birds and farmed birds, as distinct to transmission between farmed birds. It is implied but not explicitly stated that "spillover" is relevant to the transmission cycle in farmed birds only. The sentence, "we developed separate ecological niche models for wild and domestic bird HPAI occurrences ..." could have been supported by a clear sentence describing the transmission cycle, to prime the reader for why two separate models were necessary.

We respectfully disagree that the term “spillover” is unclear in the manuscript. In both the Methods and Discussion sections (lines 387-391 and 409-414), we explicitly define “spillover” as the introduction of HPAI viruses from wild birds into domestic poultry, and we distinguish this from secondary farm-to-farm transmission. Our use of separate ecological niche models for wild and domestic outbreaks reflects not only the distinction between primary spillover and secondary transmission, but also the fundamentally different ecological processes, surveillance systems, and management implications that shape outbreaks in these two groups. We will clarify this choice in the revised manuscript when introducing the separate models. Furthermore, on line 83, we will add “as these two groups are influenced by different ecological processes, surveillance biases, and management contexts”.

I also queried the importance of (dead-end) mammalian infections to a model of the HPAI transmission risk, to which the authors responded: "While spillover events of HPAI into mammals have been documented, these detections are generally considered dead-end infections and do not currently represent sustained transmission chains. As such, they fall outside the scope of our study, which focuses on avian hosts and models ecological suitability for outbreaks in wild and domestic birds." I would argue that any infections, whether they are in dead-end or competent hosts, represent the presence of environmental conditions to support transmission so are certainly relevant to a niche model and therefore within scope. It is certainly understandable if the authors have not been able to access data of mammalian infections, but it is an oversight to dismiss these infections as irrelevant.

We understand the Reviewer’s point, but our study was designed to model HPAI occurrence in avian hosts only. We therefore restricted our analysis to wild birds and domestic poultry, which represent the primary hosts for HPAI circulation and the focus of surveillance and control measures. While mammalian detections have been reported, they are outside the scope of this work.

Correlative ecological niche models, including BRTs, learn relationships between occurrence data and covariate data to make predictions, irrespective of correlations between covariates. I am not convinced that the authors can make any "interpretation" (line 298) that the covariates that are most informative to their models have any "influence" (line 282) on their response variable. Indeed, the observation that "land-use and climatic predictors do not play an important role in the niche ecological models" (line 286), while "intensive chicken population density emerges as a significant predictor" (line 282) begs the question: from an operational perspective, is the best (e.g., most interpretable and quickest to generate) model of HPAI risk a map of poultry farming intensity?

We agree that poultry density may partly reflect reporting bias, but we also assumed it a meaningful predictor of HPAI risk. Its importance in our models is therefore expected. Importantly, our BRT framework does more than reproduce poultry distribution: it captures non-linear relationships and interactions with other covariates, allowing a more nuanced characterisation of risk than a simple poultry density map. Note also that we distinguished in our models intensive and extensive chicken poultry density and duck density. Therefore, it is not a “map of poultry farming intensity”.

At line 282, we used the word “influence” while fully recognising that correlative models cannot establish causality. Indeed, in our analyses, “relative influence” refers to the importance metric produced by the BRT algorithm (Ridgeway, 2020), which measures correlative associations between environmental factors and outbreak occurrences. These scores are interpreted in light of the broader scientific literature, therefore our interpretations build on both our results and existing evidence, rather than on our models alone. However, in the next version of the paper, we will revise the sentence as: “underscoring the strong association of poultry farming practices with HPAI spread (Dhingra et al., 2016)”.

I have more significant concerns about the authors' treatment of sampling bias: "We agree with the Reviewer's comment that poultry density could have potentially been considered to guide the sampling effort of the pseudo-absences to consider when training domestic bird models. We however prefer to keep using a human population density layer as a proxy for surveillance bias to define the relative probability to sample pseudo-absence points in the different pixels of the background area considered when training our ecological niche models. Indeed, given that poultry density is precisely one of the predictors that we aim to test, considering this environmental layer for defining the relative probability to sample pseudo-absences would introduce a certain level of circularity in our analytical procedure, e.g. by artificially increasing to influence of that particular variable in our models." The authors have elected to ignore a fundamental feature of distribution modelling with occurrence-only data: if we include a source of sampling bias as a covariate and do not include it when we sample background data, then that covariate would appear to be correlated with presence. They acknowledge this later in their response to my review: "...assuming a sampling bias correlated with poultry density would result in reducing its effect as a risk factor." In other words, the apparent predictive capacity of poultry density is a function of how the authors have constructed the sampling bias for their models. A reader of the manuscript can reasonably ask the question: to what degree are is the model a model of HPAI transmission risk, and to what degree is the model a model of the observation process? The sentence at lines 474-477 is a helpful addition, however the preceding sentence, "Another approach to sampling pseudo-absences would have been to distribute them according to the density of domestic poultry," (line 474) is included without acknowledgement of the flow-on consequence to one of the key findings of the manuscript, that "...intensive chicken

population density emerges as a significant predictor..." (line 282). The additional context on the EMPRES-i dataset at line 475-476 ("the locations of outbreaks ... are often georeferenced using place name nomenclatures") is in conflict with the description of the dataset at line 407 ("precise location coordinates"). Ultimately, the choices that the authors have made are entirely defensible through a clear, concise description of model features and assumptions, and precise language to guide the reader through interpretation of results. I am not satisfied that this is provided in the revised manuscript.

We thank the Reviewer for this important point. To address it, we compared model predictive performance and covariate relative influences obtained when pseudo-absences were weighted by poultry density versus human population density (Author response table 1). The results show that differences between the two approaches are marginal, both in predictive performance (Δ AUC ranging from -0.013 to +0.002) and in the ranking of key predictors (see below Author response images 1 and 2). For instance, intensive chicken density consistently emerged as an important predictor regardless of the bias layer used.

Note: the comparison was conducted using a simplified BRT configuration for computational efficiency (fewer trees, fixed 5-fold random cross-validation, and standardised parameters). Therefore, absolute values of AUC and variable importance may differ slightly from those in the manuscript, but the relative ranking of predictors and the overall conclusions remain consistent.

Given these small differences, we retained the approach using human population density. We agree that poultry density partly reflects surveillance bias as well as true epidemiological risk, and we will clarify this in the revised manuscript by noting that the predictive role of poultry density reflects both biological processes and surveillance systems. Furthermore, on line 289, we will add "We note, however, that intensive poultry density may reflect both surveillance intensity and epidemiological risk, and its predictive role in our models should be interpreted in light of both processes".

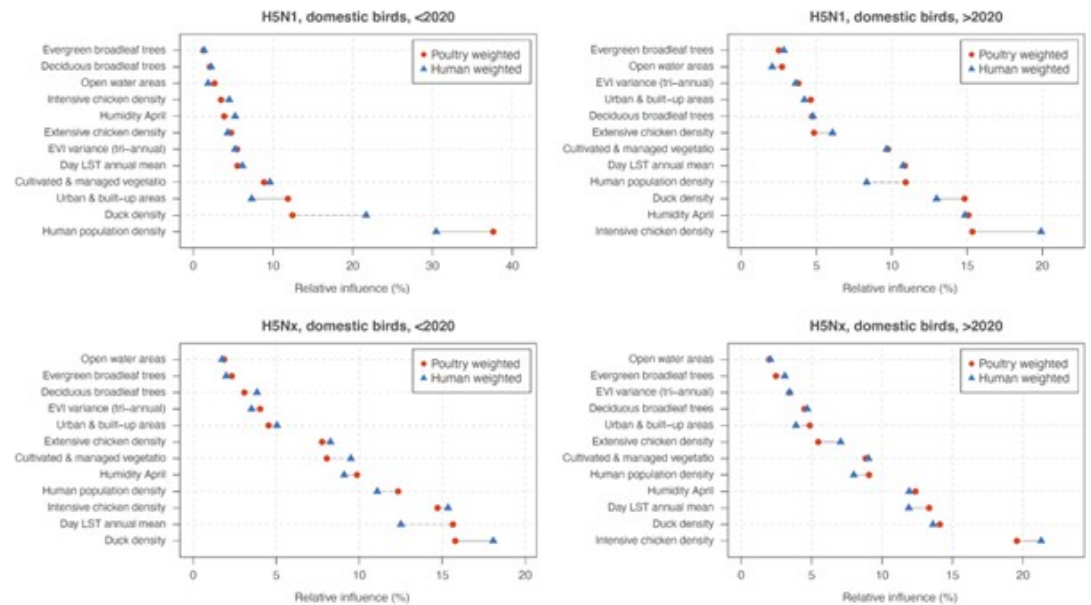
Author response table 1.

Comparison of model predictive performances (AUC) between pseudo-absence sampling were weighted by poultry density and by human population density across host groups, virus types, and time periods. Differences in AUC values are shown as the value for poultry-weighted minus human-weighted pseudo-absences.

Dataset	AUC (poultry-weighted pseudo-absences)	AUC (human-weighted pseudo-absences)	AUC differences between the two approaches
H5N1, domestic birds, <2020	0.877	0.864	-0.013
H5N1, wild birds, >2020	0.837	0.829	-0.008
H5N1, domestic birds, >2020	0.831	0.828	-0.003
H5Nx, wild birds, <2020	0.856	0.858	0.002
H5Nx, domestic birds, <2020	0.853	0.855	0.002
H5Nx, wild birds, >2020	0.847	0.842	-0.005
H5Nx, domestic birds, >2020	0.835	0.833	-0.002

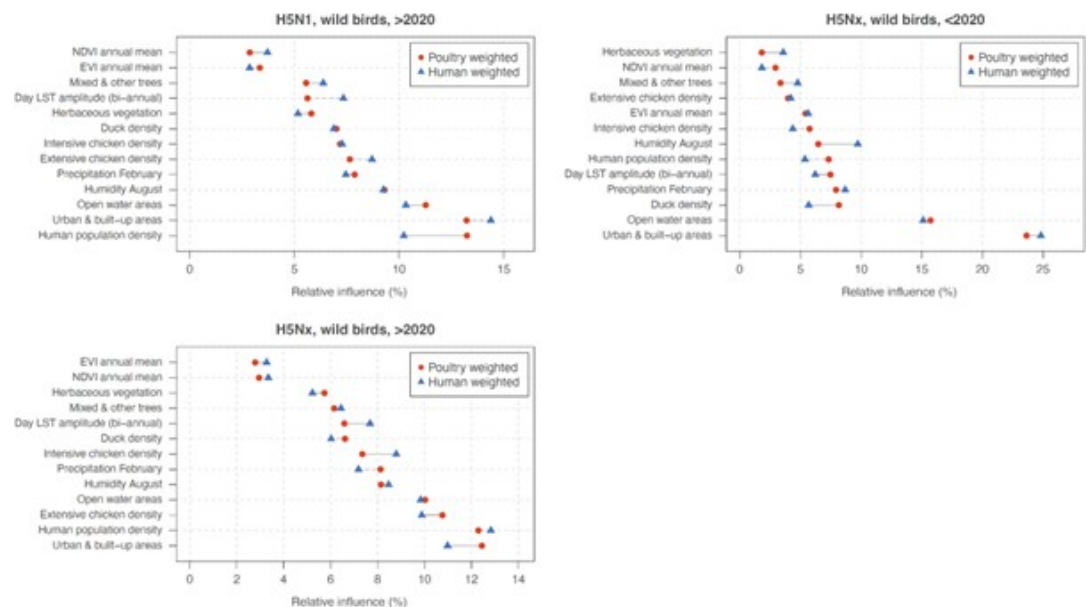
Author response image 1.

Comparison of variable relative influence (%) between models trained with pseudo-absences weighted by poultry density (red) and human population density (blue) for domestic bird outbreaks. Results are shown for four datasets: H5N1 (<2020), H5N1 (>2020), H5Nx (<2020), and H5Nx (>2020).



Author response image 2.

Comparison of variable relative influence (%) between models trained with pseudo-absences weighted by poultry density (red) and human population density (blue) for wild bird outbreaks. Results are shown for three datasets: H5N1 (>2020), H5Nx (<2020), and H5Nx (>2020).



The authors have slightly misunderstood my comment on "extrapolation": I referred to "environmental extrapolation" in my review without being particularly explicit about my meaning. By "environmental extrapolation", I meant to ask whether the models were predicting to environments that are outside the extent of environments included in the occurrence data used in the manuscript. The authors appear to have understood this to be a comment on geographic extrapolation, or predicting to areas outside the geographic extent included in occurrence data, e.g.: "For H5Nx post-2020, areas of high

predicted ecological suitability, such as Brazil, Bolivia, the Caribbean islands, and Jilin province in China, likely result from extrapolations, as these regions reported few or no outbreaks in the training data" (lines 195-197). Is the model extrapolating in environmental space in these regions? This is unclear. I do not suggest that the authors should carry out further analysis, but the multivariate environmental similarity surface (MESS; see Elith et al., 2010) is a useful tool to visualise environmental extrapolation and aid model interpretation.

On the subject of "extrapolation", I am also concerned by the additions at lines 362-370: "...our models extrapolate environmental suitability for H5Nx in wild birds in areas where few or no outbreaks have been reported. This discrepancy may be explained by limited surveillance or underreporting in those regions." The "discrepancy" cited here is a feature of the input dataset, a function of the observation distribution that should be captured in pseudo-absence data. The authors state that Kazakhstan and Central Asia are areas of interest, and that the environments in this region are outside the extent of environments captured in the occurrence dataset, although it is unclear whether "extrapolation" is informed by a quantitative tool like a MESS or judged by some other qualitative test. The authors then cite Australia as an example of a region with some predicted suitability but no HPAI outbreaks to date, however this discussion point is not linked to the idea that the presence of environmental conditions to support transmission need not imply the occurrence of transmission (as in the addition, "...spatial isolation may imply a lower risk of actual occurrences..." at line 214). Ultimately, the authors have not added any clear comment on model uncertainty (e.g., variation between replicated BRTs) as I suggested might be helpful to support their description of model predictions.

Many thanks for the clarification. Indeed, we interpreted your previous comments in terms of geographic extrapolations. We thank the Reviewer for these observations. We will adjust the wording to further clarify that predictions of ecological suitability in areas with few or no reported outbreaks (e.g., Central Asia, Australia) are not model errors but expected extrapolations, since ecological suitability does not imply confirmed transmission (for instance, on Line 362: "our models extrapolate environmental suitability" will be changed to "Interestingly, our models extrapolate geographical"). These predictions indicate potential environments favorable to circulation if the virus were introduced.

In our study, model uncertainty is formally assessed when comparing the predictive performances of our models (Fig. S3, Table S1), the relative influence (Table S3) and response curves (Fig. 2) associated with each environmental factor (Table S2). All the results confirming a good converge between these replicates. Finally, we indeed did not use a quantitative tool such as a MESS to assess extrapolation but did rely on qualitative interpretation of model outputs.

All of my criticisms are, of course, applied with the understanding that niche modelling is imperfect for a disease like HPAI, and that data may be biased/incomplete, etc.: these caveats are common across the niche modelling literature. However, if language around the transmission cycle, the niche, and the interpretation of any of the models is imprecise, which I find it to be in the revised manuscript, it undermines all of the science that is presented in this work.

We respectfully disagree with this comment. The scope of our study and the methods employed are clearly defined in the manuscript, and the limitations of ecological niche modelling in this context are explicitly acknowledged in the Discussion section. While we appreciate the Reviewer's concern, the comment does not provide specific examples of unclear or imprecise language regarding the transmission cycle, niche, or interpretation of the models. Without such examples, it is difficult to identify further revisions that would improve clarity.

Reviewer #2 (Public review):

The geographic range of highly pathogenic avian influenza cases changed substantially around the period 2020, and there is much interest in understanding why. Since 2020 the pathogen irrupted in the Americas and the distribution in Asia changed dramatically. This study aimed to determine which spatial factors (environmental, agronomic and socio-economic) explain the change in numbers and locations of cases reported since 2020 (2020--2023). That's a causal question which they address by applying correlative environmental niche modelling (ENM) approach to the avian influenza case data before (2015--2020) and after 2020 (2020--2023) and separately for confirmed cases in wild and domestic birds. To address their questions they compare the outputs of the respective models, and those of the first global model of the HPAI niche published by Dhingra et al 2016.

We do not agree with this comment. In the manuscript, it is well established that we are quantitatively assessing factors that are associated with occurrences data before and after 2020. We do not claim to determine the causality. One sentence of the Introduction section (lines 75-76) could be confusing, so we intend to modify it in the final revision of our manuscript.

ENM is a correlative approach useful for extrapolating understandings based on sparse geographically referenced observational data over un- or under-sampled areas with similar environmental characteristics in the form of a continuous map. In this case, because the selected covariates about land cover, use, population and environment are broadly available over the entire world, modelled associations between the response and those covariates can be projected (predicted) back to space in the form of a continuous map of the HPAI niche for the entire world.

We fully agree with this assessment of ENM approaches.

Strengths:

The authors are clear about expected bias in the detection of cases, such geographic variation in surveillance effort (testing of symptomatic or dead wildlife, testing domestic flocks) and in general more detections near areas of higher human population density (because if a tree falls in a forest and there is no-one there, etc), and take steps to ameliorate those. The authors use boosted regression trees to implement the ENM, which typically feature among the best performing models for this application (also known as habitat suitability models). They ran replicate sets of the analysis for each of their model targets (wild/domestic x pathogen variant), which can help produce stable predictions. Their code and data is provided, though I did not verify that the work was reproducible.

The paper can be read as a partial update to the first global model of H5Nx transmission by Dhingra and others published in 2016 and explicitly follows many methodological elements. Because they use the same covariate sets as used by Dhingra et al 2016 (including the comparisons of the performance of the sets in spatial cross-validation) and for both time periods of interest in the current work, comparison of model outputs is possible. The authors further facilitate those comparisons with clear graphics and supplementary analyses and presentation. The models can also be explored interactively at a weblink provided in text, though it would be good to see the model training data there too.

The authors' comparison of ENM model outputs generated from the distinct HPAI case datasets is interesting and worthwhile, though for me, only as a response to differently

framed research questions.

Weaknesses:

This well-presented and technically well-executed paper has one major weakness to my mind. I don't believe that ENM models were an appropriate tool to address their stated goal, which was to identify the factors that "explain" changing HPAI epidemiology.

Here is how I understand and unpack that weakness:

(1) Because of their fundamentally correlative nature, ENMs are not a strong candidate for exploring or inferring causal relationships.

(2) Generating ENMs for a species whose distribution is undergoing broad scale range change is complicated and requires particular caution and nuance in interpretation (e.g., Elith et al, 2010, an important general assumption of environmental niche models is that the target species is at some kind of distributional equilibrium (at time scales relevant to the model application). In practice that means the species has had an opportunity to reach all suitable habitats and therefore its absence from some can be interpreted as either unfavourable environment or interactions with other species). Here data sets for the response (N5H1 or N5Hx case data in domestic or wild birds) were divided into two periods; 2015--2020, and 2020--2023 based on the rationale that the geographic locations and host-species profile of cases detected in the latter period was suggestive of changed epidemiology. In comparing outputs from multiple ENMs for the same target from distinct time periods the authors are expertly working in, or even dancing around, what is a known grey area, and they need to make the necessary assumptions and caveats obvious to readers.

We thank the Reviewer for this observation. First, we constrained pseudo-absence sampling to countries and regions where outbreaks had been reported, reducing the risk of interpreting non-affected areas as environmentally unsuitable. Second, we deliberately split the outbreak data into two periods (2015-2020 and 2020-2023) because we do not assume a single stable equilibrium across the full study timeframe. This division reflects known epidemiological changes around 2020 and allows each period to be modeled independently. Within each period, ENM outputs are interpreted as associations between outbreaks and covariates, not as equilibrium distributions. Finally, by testing prediction across periods, we assessed both niche stability and potential niche shifts. These clarifications will be added to the manuscript to make our assumptions and limitations explicit.

Line 66, we will add: "Ecological niche model outputs for range-shifting pathogens must therefore be interpreted with caution (Elith et al., 2010). Despite this limitation, correlative ecological niche models remain useful for identifying broad-scale associations and potential shifts in distribution. To account for this, we analysed two distinct time periods (2015-2020 and 2020-2023)."

Line 123, we will revise "These findings underscore the ability of pre-2020 models in forecasting the recent geographic distribution of ecological suitability for H5Nx and H5N1 occurrences" to "These results suggest that pre-2020 models captured broad patterns of suitability for H5Nx and H5N1 outbreaks, while post-2020 models provided a closer fit to the more recent epidemiological situation".

(3) To generate global prediction maps via ENM, only variables that exist at appropriate resolution over the desired area can be supplied as covariates. What processes could influence changing epidemiology of a pathogen and are their covariates that represent them? Introduction to a new geographic area (continent) with naive population, immunity in previously exposed populations, control measures to limit spread such as

vaccination or destruction of vulnerable populations or flocks? Might those control measures be more or less likely depending on the country as a function of its resources and governance? There aren't globally available datasets that speak to those factors, so the question is not why were they omitted but rather was the authors decision to choose ENMs given their question justified? How valuable are insights based on patterns of correlation change when considering different temporal sets of HPAI cases in relation to a common and somewhat anachronistic set of covariates?

We agree that the ecological niche models trained in our study are limited to environmental and host factors, as described in the Methods section with the selection of predictors. While such models cannot capture causality or represent processes such as immunity, control measures, or governance, they remain a useful tool for identifying broad associations between outbreak occurrence and environmental context. Our study cannot infer the full mechanisms driving changes in HPAI epidemiology, but it does provide a globally consistent framework to examine how associations with available covariates vary across time periods.

*(4) In general the study is somewhat incoherent with respect to time. Though the case data come from different time periods, each response dataset was modelled separately using exactly the same covariate dataset that predated both sets. That decision should be understood as a strong assumption on the part of the authors that conditions the interpretation: the world (as represented by the covariate set) is immutable, so the model has to return different correlative associations between the case data and the covariates to explain the new data. While the world represented by the selected covariates *may* be relatively stable (could be statistically confirmed), what about the world not represented by the covariates (see point 3)?*

We used the same covariate layers for both periods, which indeed assumes that these environmental and host factors are relatively stable at the global scale over the short timeframe considered. We believe this assumption is reasonable, as poultry density, land cover, and climate baselines do not change drastically between 2015 and 2023 at the resolution of our analysis. We agree, however, that unmeasured processes such as control measures, immunity, or governance may have changed during this time and are not captured by our covariates.

Recommendations for the Authors:

Reviewer #1 (Recommendations for the authors):

- Line 400-401: "over the 2003-2016 periods" has an extra "s"; "two host species" (with reference to wild and domestic birds) would be more precise as "two host groups".

- Remove comma line 404

Many thanks for these comments, we have modified the text accordingly.

Reviewer #2 (Recommendations for the authors):

Most of my work this round is encapsulated in the public part of the review.

The authors responded positively to the review efforts from the previous round, but I was underwhelmed with the changes to the text that resulted. Particularly in regard to limiting assumptions - the way that they augmented the text to refer to limitations raised in review downplayed the importance of the assumptions they've made. So they acknowledge the significance of the limitation in their rejoinder, but in the amended text merely note the limitation without giving any sense of what it means for their interpretation of the findings of this study.

The abstract and findings are essentially unchanged from the previous draft.

I still feel the near causal statements of interpretation about the covariates are concerning. These models really are not a good candidate for supporting the inference that they are making and there seem to be very strong arguments in favour of adding covariates that are not globally available.

We never claimed causal interpretation, and we have consistently framed our analyses in terms of associations rather than mechanisms. We acknowledge that one phrasing in the research questions (“Which factors can explain...”) could be misinterpreted, and we are correcting this in the revised version to read “Which factors are associated with...”. Our approach follows standard ecological niche modelling practice, which identifies statistical associations between occurrence data and covariates. As noted in the Discussion section, these associations should not be interpreted as direct causal mechanisms. Finally, all interpretive points in the manuscript are supported by published literature, and we consider this framing both appropriate and consistent with best practice in ecological niche modelling (ENM) studies.

We assessed predictor contributions using the “relative influence” metric, the terminology reported by the R package “gbm” (Ridgeway, 2020). This metric quantifies the contribution of each variable to model fit across all trees, rescaled to sum to 100%, and should be interpreted as an association rather than a causal effect.

L65-66 The general difficulty of interpreting ENM output with range-shifting species should be cited here to alert readers that they should not blithely attempt what follows at home.

I believe that their analysis is interesting and technically very well executed, so it has been a disappointment and hard work to write this assessment. My rough-cut last paragraph of a reframed intro would go something like - there are many reasons in the literature not to do what we are about to do, but here's why we think it can be instructive and informative, within certain guardrails.

To acknowledge this comment and the previous one, we revised lines 65-66 to: “However, recent outbreaks raise questions about whether earlier ecological niche models still accurately predict the current distribution of areas ecologically suitable for the local circulation of HPAI H5 viruses. Ecological niche model outputs for range-shifting pathogens must therefore be interpreted with caution (Elith et al., 2010). Despite this limitation, correlative ecological niche models remain useful for identifying broad-scale associations and potential shifts in distribution.”

We respectfully disagree with the Reviewer’s statement that “there are many reasons in the literature not to do what we are about to do”. All modeling approaches, including mechanistic ones, have limitations, and the literature is clear on both the strengths and constraints of ecological niche models. Our manuscript openly acknowledges these limits and frames our findings accordingly. We therefore believe that our use of an ENM approach is justified and contributes valuable insights within these well-defined boundaries.

Reference: Ridgeway, G. (2007). Generalized Boosted Models: A guide to the gbm package. Update, 1(1), 2007.

The following is the authors’ response to the original reviews.

Reviewer #1(Public review):

I am concerned by the authors' conceptualisation of "niche" within the manuscript. Is the "niche" we are modelling the niche of the pathogen itself? The niche of the (wild) bird host species as a group? The niche of HPAI transmission within (wild) bird host species (i.e., an intersection of pathogen and bird niches)? Or the niche of HPAI transmission in poultry? The precise niche being modelled should be clarified in the Introduction or early in the Methods of the manuscript. The first two definitions of niche listed above are relevant, but separate from the niche modelled in the manuscript - this should be acknowledged.

We acknowledge that these concepts were probably not enough clearly defined in the previous version of our manuscript, and we have now included an explicit definition in the fourth paragraph of the Introduction section: “We developed separate ecological niche models for wild and domestic bird HPAI occurrences, these models thus predicting the ecological suitability for the risk of local viral circulation leading to the detection of HPAI occurrences within each host group (rather than the niche of the virus or the host species alone).”

The authors should consider the precise transmission cycle involved in each HPAI case: "index cases" in farmed poultry, caused by "spillover" from wild birds, are relevant to the wildlife transmission cycle, while the ecological conditions coinciding with subsequent transmission in farmed poultry are likely to be fundamentally different. (For example, subsequent transmission is not conditional on the presence of wild birds.) Modelling these two separate, but linked, transmission cycles together may omit important nuances from the modelling framework.

We thank the Reviewer for highlighting the distinction between primary (wild-to-domestic) and secondary (farm-to-farm) transmission cycles. Our modelling framework was designed to assess the ecological suitability of HPAI occurrences in wild and domestic birds separately. In the domestic poultry models, the response variables are the confirmed outbreaks data and do not distinguish between index cases resulting from primary or secondary infections.

One of the aims of the study is to evaluate the spatial distribution of areas ecologically suitable for local H5N1/x circulation either leading to domestic or wild bird cases, i.e. to identify environmental conditions where the virus may have persisted or spread, whether as a result of introduction by wild birds or farm-to-farm transmission. Introducing mechanistic distinctions in the response variable would not necessarily improve or affect the ecological suitability maps, since each type of transmission is likely to be associated with different covariates that are included in the models.

Also, the EMPRES-i database does not indicate whether each record corresponds to an index case or a secondary transmission event, so in practice it would not be possible to produce two different models. However, we agree that distinguishing between types of transmission is an interesting perspective for future research. This could be explored, for example, by mapping interfaces between wild and domestic bird populations or by inferring outbreak transmission trees using genomic data when available.

To avoid confusion, we now explicitly clarify this aspect in the Materials and Methods section: “It is important to note that the EMPRES-i database does not distinguish between index cases (e.g., primary spillover from wild birds) and secondary farm-to-farm transmissions. As such, our ecological niche models are trained on confirmed HPAI outbreaks in poultry that may result from different transmission dynamics — including both initial introduction events influenced by environmental factors and subsequent spread within poultry systems.”

We now also address this limitation in the Discussion section: “Finally, our models for domestic poultry do not distinguish between primary introduction events (e.g., spillover from wild birds) and secondary transmission between farms due to limitations in the available surveillance data. While environmental factors likely influence the risk of initial spillover events, secondary spread is more often driven by anthropogenic factors such as biosecurity practices and poultry trade, which are not included in our current modelling framework.”

The authors should clarify the meaning of "spillover" within the HPAI transmission cycle: if spillover transmission is from wild birds to farmed poultry, then subsequent transmission in poultry is separate from the wildlife transmission cycle. This is particularly relevant to the Discussion paragraph beginning at line 244: does "farm to farm transmission" have a distinct ecological niche to transmission between wild birds, and transmission between wild birds and farmed birds? And while there has been a spillover of HPAI to mammals, could the authors clarify that these detections are dead-end? And not represented in the dataset? Dhingra et al., 2016 comment on the contrast between models of "directly transmitted" pathogens, such as HPAI, and vector-borne diseases: for vector-borne diseases, "clear eco-climatic boundaries of vectors can be mapped", whereas "HPAI is probably not as strongly environmentally constrained". This is an important piece of nuance in their Discussion and a comment to a similar effect may be of use in this manuscript.

Following the Reviewer’s previous comment, we have now added clarifications in the Methods and Discussion sections defining spillover as the transmission of HPAI viruses from wild birds to domestic poultry (index cases), and secondary transmission as onward spread between farms. As mentioned in our answer above, we now emphasise that our models do not distinguish these dynamics, which are likely to be influenced by different drivers — ecological in the case of spillover, and often anthropogenic (e.g., poultry trade movement, biosecurity) in the case of farm-to-farm transmission.

The discussion regarding farm-to-farm transmission and spillovers is indeed an interpretation derived from the covariates analysis (see the second paragraph in the Discussion section). Specifically, we observed a stronger association between HPAI occurrences and domestic bird density after 2020, which may suggest that secondary infections (e.g., farm-to-farm transmission) became more prominent or more frequently reported. We however acknowledge that our data do not allow us to distinguish primary introductions from secondary transmission events, and we have added a sentence to explicitly clarify this: “However, this remains an interpretation, as the available data do not allow us to distinguish between index cases and secondary transmission events.”

We thank the Reviewer for raising the point of mammalian infections. While spillover events of HPAI into mammals have been documented, these detections are generally considered dead-end infections and do not currently represent sustained transmission chains. As such, they fall outside the scope of our study, which focuses on avian hosts and models ecological suitability for outbreaks in wild and domestic birds. However, we agree that future work could explore the spatial overlap between mammalian outbreak detections and ecological suitability maps for wild birds to assess whether such spillovers may be linked to localised avian transmission dynamics.

Finally, we have added a comment about the differences between pathogens strongly constrained by the environments and HPAI: “This suggests that HPAI H5Nx is not as strongly environmentally constrained as vector-borne pathogens, for which clear eco-climatic boundaries (e.g., vector borne diseases) can be mapped (Dhingra et al., 2016).” This aligns with the interpretation provided by Dhingra and colleagues (2016) and helps contextualise

the predictive limitations of ecological niche models for directly transmitted pathogens like HPAI.

There are several places where some simple clarification of language could answer my questions related to ecological niches. For example, on line 74, "the ecological niche" should be followed by "of the pathogen", or "of HPAI transmission in wild birds", or some other qualifier that is most appropriate to the Authors' conceptualisation of the niche modelled in the manuscript. Similarly, in the following sentence, "areas at risk" could be followed by "of transmission in wild birds", to make the transmission cycle that is the subject of modelling clear to the reader. On line 83, it is not clear who or what is the owner of "their ecological niches": is this "poultry and wild birds", or the pathogen?

We agree with that suggestion and have now modified the related part of the text accordingly (e.g., “areas at risk for local HPAI circulation” and “of HPAI in wild or domestic birds”).

I am concerned by the authors' treatment of sampling bias in their BRT modelling framework. If we are modelling the niche of HPAI transmission, we would expect places that are more likely to be subject to disease surveillance to be represented in the set of locations where the disease has been detected. I do not agree that pseudo-absence points are sampled "to account for the lack of virus detection in some areas" - this description is misleading and does not match the following sentence ("pseudo-absence points sampled ... to reflect the greater surveillance efforts ..."). The distribution of pseudo-absences should aim to capture the distribution of probable disease surveillance, as these data act as a stand-in for missing negative surveillance records. It is sensible that pseudo-absences for disease detection in wild birds are sampled proportionately to human population density, as the disease is detected in dead wild birds, which are more likely to be identified close to areas of human occupation (as stated on line 163). However, I do not agree that the same applies to poultry - the density of farmed poultry is likely to be a better proxy for surveillance intensity in farmed birds. Human population density and farmed poultry density may be somewhat correlated (i.e., both are low in remote areas), but poultry density is likely to be higher in rural areas, which are assumed to have relatively lower surveillance intensity under the current approach. The authors allude to this in the Discussion: "monitoring areas with high intensive chicken densities ... remains crucial for the early detection and management of HPAI outbreaks".

We agree with the Reviewer's comment that poultry density could have potentially been considered to guide the sampling effort of the pseudo-absences to consider when training domestic bird models. We however prefer to keep using a human population density layer as a proxy for surveillance bias to define the relative probability to sample pseudoabsence points in the different pixels of the background area considered when training our ecological niche models. Indeed, given that poultry density is precisely one of the predictors that we aim to test, considering this environmental layer for defining the relative probability to sample pseudo-absences would introduce a certain level of circularity in our analytical procedure, e.g. by artificially increasing to influence of that particular variable in our models.

Furthermore, it is also worth noting that, to better account for variations in surveillance intensity, we also adjusted the sampling effort by allocating pseudo-absences in proportion to the number of confirmed outbreaks per administrative unit (country or sub-national regions for Russia and China). This approach aimed to reduce bias caused by uneven reporting and surveillance efforts between regions. Additionally, we restricted model training to countries or regions with a minimum surveillance threshold (at least five confirmed outbreaks per administrative unit). Therefore, both presence and pseudo-absence points originated from areas with more consistent surveillance data.

We acknowledge in the Materials and Methods section that the approach proposed by the Reviewer could have been used: “Another approach to sampling pseudo-absences would have been to distribute them according to the density of domestic poultry.” Finally, our approach is also justified in our response to the next comment of the Reviewer.

Having written my review, including the paragraph above, I briefly scanned Dhingra et al., and found that they provide justification for the use of human population density to sample pseudoabsences in farmed birds: “the Empres-i database compiles outbreak locations data from very heterogeneous sources and in the absence of explicit GPS location data, the geo-referencing of individual cases is often through the use of place name gazetteers that will tend to force the outbreak location populated place, rather in the exact location of the farm where the disease was found, which would introduce a bias correlated with human population density.” This context is entirely missing from the manuscript under review, however, I maintain the comment in the paragraph above - have the Authors trialled sampling pseudo-absences from poultry density layers?

We agree with the Reviewer’s comment and have now added this precision in the Materials and Methods section (in the third paragraph dedicated to ecological niche modelling): “However, as pointed out by Dhingra and colleagues (2016), the locations of outbreaks in the EMPRES-i database are often georeferenced using place name nomenclatures due to a lack of accurate GPS data, which could introduce a spatial bias towards populated areas.”

The authors indirectly acknowledge the role of sampling bias in model predictions at line 163, however, this point could be clearer: there is sampling bias in the set of locations where HPAI has been observed and failure to adequately replicate this sampling bias in pseudo-absence data could lead covariates that are correlated with the observation distribution to appear to be correlated with the target distribution. This point is alluded to but should be clearly acknowledged to allow the reader to appropriately interpret your results. I understand the point being made on line 163 is that surveillance of HPAI in wild birds has become more structured and less opportunistic over time - if this is the case, a statement to this effect could replace “which could influence earlier data sets”, which is a little ambiguous. The Authors acknowledge the role of sampling bias in lines 241-242 - this may be a good place to remind the reader that they have attempted to incorporate sampling bias through the selection of their pseudoabsence dataset, particularly for wild bird models.

We thank the Reviewer for this comment. We have now clarified in the text that observed data on HPAI occurrence are inherently influenced by heterogeneous surveillance efforts and that failure to replicate this bias in pseudo-absence sampling could effectively lead to misleading correlations with covariates associated with surveillance effort rather than true ecological suitability. We have now rephrased the related sentence as follows: “This decline may indicate a reduced bias in observation data: typically, dead wild birds are more frequently found near human-populated areas due to opportunistic detections, whereas more recent surveillance efforts have become increasingly proactive (Giacinti et al., 2024).”

Dhingra et al. aimed to account for the effect of mass vaccination of birds in China. This does not appear to be included in the updated models - is this a relevant covariate to consider in updated models? Are the models trained on pre-2020 data predicting to post-2020 given the same presence dataset as previous models? It may be helpful to provide a comment on this if we consider the pre-2020 models in this work to be representative of pre-2020 models as a cohort. Given the framing of the manuscript as an update to Dhingra et al., it may be useful for the authors to briefly summarise any differences between the existing models and updated models. Dhingra et al., also examine spatial

extrapolation, which is not addressed here. Environmental extrapolation may be a useful metric to consider: are there areas where models are extrapolating that are predicted to be at high risk of HPAI transmission? Finally, they also provide some inset panels on global maps of model predictions - something similar here may also be useful.

We thank the Reviewer for these comments. Vaccination coverage is indeed a relevant covariate for HPAI suitability in domestic birds. However, we did not include this variable in our updated models for two reasons. First, comprehensive vaccination data were only available for China, so it is not possible to include this variable in a global model. Second, available data were outdated and vaccination strategies can vary substantially over time.

We however agree with the Reviewer that the Materials and Methods section did not clarify clearly the differences with Dhingra et al. (2016), and we now detail these differences at the beginning of the Materials and Methods section: “Our approach is similar to the one implemented by Dhingra and colleagues (2016). While Dhingra et al. (2016) developed their models only for domestic birds over the 2003-2016 periods, our models were developed for two host species separately (wild and domestic birds) and for two time periods (2016-2020 and 2020-2023).”

We also detail the main difference concerning the pseudo-absences sampling: Dhingra and colleagues (2016) used human population density to sample pseudo-absences to reflect potential surveillance bias and also account for spatial filtering (min/max distances from presence). We adopted a similar strategy but also incorporated outbreak count per country or province (in the case of China and Russia) into the pseudo-absence sampling process to further account for within-country surveillance heterogeneity. We have now added these specifications in the Materials and Methods section: “To account for heterogeneity in AIV surveillance and minimise the risk of sampling pseudo-absences in poorly monitored regions, we restricted our analysis to countries (or administrative level 1 units in China and Russia) with at least five confirmed outbreaks. Unlike Dhingra et al. (2016), who sampled pseudoabsences across a broader global extent, our sampling was limited to regions with demonstrated surveillance activity. In addition, we adjusted the density of pseudo-absence points according to the number of reported outbreaks in each country or admin-1 unit, as a proxy for surveillance effort — an approach not implemented in this previous study.”

We have now also provided a comparison between the different outputs, particularly in the Results section: “Our findings were overall consistent with those previously reported by Dhingra and colleagues (Dhingra et al., 2016), who used data from January 2004 to March 2015 for domestic poultry. However, some differences were noted: their maps identified higher ecological suitability for H5 occurrences before 2016 in North America, West Africa, eastern Europe, and Bangladesh, while our maps mainly highlight ecologically suitable regions in China, South-East Asia, and Europe (Fig. S5). In India, analyses consistently identified high ecologically suitable areas for the risk of local H5Nx and H5N1 circulation for the three time periods (pre-2016, 2016-2020, and post-2020). Similar to the results reported by Dhingra and colleagues, we observed an increase in the ecological suitability estimated for H5N1 occurrence in South America's domestic bird populations post-2020. Finally, Dhingra and colleagues identified high suitability areas for H5Nx occurrence in North America, which are predicted to be associated with a low ecological suitability in the 2016-2020 models.”

We acknowledge that some regions predicted as highly suitable correspond to areas where extrapolation likely occurs due to limited or no recorded outbreaks. We have now added these specifications when discussing the resulting suitability maps obtained for domestic birds: “For H5Nx post-2020, areas of high predicted ecological suitability, such as Brazil, Bolivia, the Caribbean islands, and Jilin province in China, likely result from extrapolations, as these regions reported few or no outbreaks in the training data”, and, for wild birds: “Some of the areas with high predicted ecological suitability reflect the result of

extrapolations. This is particularly the case in coastal regions of West and North Africa, the Nile Basin, Central Asia (Kyrgyzstan, Tajikistan, Uzbekistan), Brazil (including the Amazon and coastal areas), southern Australia, and the Caribbean, where ecological conditions are similar to those in areas where outbreaks are known to occur but where records of outbreaks are still rare.”

For wild birds (H5Nx, post-2020), high ecological suitability was predicted along the West and North African coasts, the Nile basin, Central Asia (e.g., Kyrgyzstan, Tajikistan, Uzbekistan), the Brazilian coast and Amazon region, Caribbean islands, southern Australia, and parts of Southeast Asia. Ecological suitability estimated in these regions may directly result from extrapolations and should therefore be interpreted cautiously.

We also added a discussion of the extrapolation for wild birds (in the Discussion section): “Interestingly, our models extrapolate environmental suitability for H5Nx in wild birds in areas where few or no outbreaks have been reported. This discrepancy may be explained by limited surveillance or underreporting in those regions. For instance, there is significant evidence that Kazakhstan and Central Asia play a role as a centre for the transmission of avian influenza viruses through migratory birds (Amirgazin et al., 2022; FAO, 2005; Sultankulova et al., 2024). However, very few wild bird cases are reported in EMPRES-i. In contrast, Australia appears environmentally suitable in our models, yet no incursion of HPAI H5N1 2.3.4.4b has occurred despite the arrival of millions of migratory shorebirds and seabirds from Asia and North America. Extensive surveillance in 2022 and 2023 found no active infections nor evidence of prior exposure to the 2.3.4.4b lineage (Wille et al., 2024; Wille and Klaassen, 2023).”

We agree that inset panels can be helpful for visualising global patterns. However, all resulting maps are available on the MOOD platform (<https://app.mood-h2020.eu/core>), which provides an interactive interface allowing users to zoom in and out, identify specific locations using a background map, and explore the results in greater detail. This resource is referenced in the manuscript to guide readers to the platform.

Related to my review of the manuscript's conceptualisation above, there are several inconsistencies in terminology in the manuscript - clearing these up may help to make the methods and their justification clearer to the reader. The "signal" that the models are estimating is variously described as "susceptibility" and "risk" (lines 179-180), "HPAI H5 ecological suitability" (line 78), "likelihood of HPAI occurrences" (line 139), "risk of HPAI circulation" (line 187), "distribution of occurrence data" (line 428). Each of these quantities has slightly different meanings and it is confusing to the reader that all of these descriptors are used for model output. "Likelihood of HPAI occurrences" is particularly misleading: ecological niche models predict high suitability for a species in areas that are similar to environments where it has previously been identified, without imposing constraints on species movement. It is intuitively far more likely that there will be HPAI occurrences in areas where the disease is already established than in areas where an introduction event is required, however, the niche models in this work do not include spatial relationships in their predictions.

We agree with the Reviewer’s comments. We have now modified the text so that in the Results section we refer to ecological suitability when referring to the outputs of the models. In the context of our Discussion section, we then interpret this ecological suitability in terms of risk, as areas with high ecological suitability being more likely to support local HPAI outbreaks.

I also caution the authors in their interpretation of the results of BRTs, which are correlative models, so therefore do not tell us what causes a response variable, but rather what is correlated with it. On Line 31, "correlated with" may be more appropriate

than "influenced by". On Line 82, "correlated with" is more appropriate than "driving". This is particularly true given the authors' treatment of sampling bias.

We agree with the Reviewer's comment and have now rephrased these sentences as follows: "The spatial distribution of HPAI H5 occurrences in wild birds appears to be primarily correlated with urban areas and open water regions" and "Our results provide a better understanding of HPAI dynamics by identifying key environmental factors correlated with the increase in H5Nx and H5N1 cases in poultry and wild birds, investigating potential shifts in their ecological niches, and improving the prediction of at-risk areas."

The following sentences in line 201 are ambiguous: "For both H5Nx and H5N1, however, isolated areas on the risk map should be interpreted with caution. These isolated areas may result from sparse data, model limitations, or local environmental conditions that may not accurately reflect true ecological suitability." By "isolated", do the authors mean remote? Or ecologically dissimilar from the set of locations where HPAI has been detected? Or ecologically dissimilar from the set of locations in the joint set of HPAI detection locations and pseudo-absences? Or ecologically similar to the set of locations where HPAI has been detected but spatially isolated? These four descriptors are each slightly different and change the meaning of the sentences. "Model limitations" are also ambiguous - could the authors clarify which specific model limitations they are referring to here? Ultimately, the point being made is probably that a model may predict high ecological suitability for HPAI transmission in areas where the disease has not yet been identified, or where a model is extrapolating in environmental space, however, uncertainty in these predictions may be greater than uncertainty in predictions in areas that are represented in surveillance data. A clear comment on model uncertainty and how it is related to the surveillance dataset and the covariate dataset is currently missing from the manuscript and would be appropriate in this paragraph.

We understand the Reviewer's concerns regarding these potential ambiguities, and have now rephrased these sentences as follows: "For both H5Nx and H5N1, certain areas of predicted high ecological suitability appear spatially isolated, i.e. surrounded by regions of low predicted ecological suitability. These areas likely meet the environmental conditions associated with past HPAI occurrences, but their spatial isolation may imply a lower risk of actual occurrences, particularly in the absence of nearby outbreaks or relevant wild bird movements."

I am concerned by the wording of the following sentence: "The risk maps reveal that high-risk areas have expanded after 2020" (line 203). This statement could be supported by an acknowledgement of the assumptions the models make of the HPAI niche: are we saying that the niche is unchanged in environmental space and that there are now more geographic areas accessible to the pathogen, or that the niche has shifted or expanded, and that there are now more geographic areas accessible to the pathogen? The authors should review the sentence beginning on line 117: if models trained on data from the old timepoint predicting to the new timepoint are almost as good as models trained on data from the new timepoint predicting to the new timepoint, doesn't this indicate that the niche, as the models are able to capture it, has not changed too much?

We thank the Reviewer for this comment. The statement that "high-risk areas have expanded after 2020" indeed refers to an increase in the geographic extent of areas predicted to have high ecological suitability in models trained on post-2020 data. This expansion likely reflects new outbreak data from regions that had not previously reported cases, which in turn influenced model training.

However, models trained on pre-2020 data retain reasonable predictive performance when applied to post-2020 data (see the AUC results reported in Table S1), suggesting that the models suggest an expansion in the ecological suitability, but do not provide definitive evidence of a shift in the ecological niche. We have now added a statement at the end of this paragraph to clarify this point: “However, models trained on pre-2020 data maintained reasonable predictive performance when tested on post-2020 data, suggesting that the overall ecological niche of HPAI did not drastically shift over time.”

The final two paragraphs of the Results might be more helpful to include at the beginning of the Results, as the data discussed there are inputs to the models. Is it possible that the "rise in Shannon index for sea birds" that "suggests a broadening of species diversity within this category from 2020 onwards" is caused by the increasingly structured surveillance of HPAI in wild birds alluded to earlier in the Results? Is the "prevalence" discussed in line 226 the frequency of the families Laridae and Sulidae being represented in HPAI detection data? Or the abundance of the bird species themselves? The language here is a little ambiguous. Discussion of particular values of Shannon/Simpson indices is slightly out of context as the meanings of the indices are in the Methods - perhaps a brief explanation of the uses of Shannon/Simpson indices may be helpful to the reader here. It may also be helpful to readers who are not acquainted with avian taxonomy to provide common names next to formal names (for example, in brackets) in the body of the text, as this manuscript is published in an interdisciplinary journal.

We thank the Reviewer for these comments. First, we acknowledge that the paragraphs on species diversity and Shannon/Simpson indices describe important data, but we have chosen to present them after the main modelling results in order to maintain a logical narrative flow. Our manuscript first presents the ecological niche models and their predictive performance, followed by interpretations of the observed patterns, including changes in avian host diversity. Diversity indices were used primarily to support and contextualise the patterns observed in the modelling results.

For clarity, we have revised the relevant paragraphs in the Results (i) to briefly remind readers of the interpretation of the Shannon and Simpson indices (“Note that these indices reflect the diversity of bird species detected in outbreak records, not necessarily their abundance in the wild”) and (ii) to clarify that “prevalence” refers to the frequency of HPAI detection in wild bird species of the Laridae (gulls) and Sulidae (boobies and gannets) families, and not their total abundance. Family of birds includes several species, so the “common name” of a family can sometimes refer to species from other families. We have now added the common names for each family in the manuscript (even if we indeed acknowledge that “penguins” can be ambiguous).

In the Methods, it is stated: "To address the heterogeneity of AIV surveillance efforts and to avoid misclassifying low-surveillance areas as unsuitable for virus circulation, we trained the ecological niche models only considering countries in which five or more cases have been confirmed." However, it is not clear how this processing step prevents low-surveillance areas from being misclassified. If pseudo-absences are appropriately sampled, low-surveillance areas should be less represented in the pseudo-absence dataset, which should lead the models to be uncertain in their predictions of these areas. Perhaps "To address the heterogeneity of AIV surveillance efforts and to avoid sampling pseudo-absence data in realistically low-surveillance areas" is a more accurate introduction to the paragraph. I am not entirely convinced that it is appropriate to remove detection data where the national number of cases is low. This may introduce further sampling bias into the dataset.

We take the opportunity of the Reviewer's comment to further clarify this important step aiming to mitigate bias associated with countries with substantial uncertainty in reporting and/or potentially insufficient HPAI surveillance data. While we indeed acknowledge that this procedure may exclude countries that had effective surveillance but low virus detection, we argue that it constitutes a relevant conservative approach to minimising the risk of sampling a significant number of pseudo-absence points in areas associated with relatively high yet undetected local HPAI circulation due to insufficient surveillance. Furthermore, given that five cases over two decades is a relatively low threshold — particularly for a highly transmissible virus such as AIV — non-detection or non-reporting remains a more plausible explanation than true absence.

To improve clarity, we have now revised the related sentence as follows: “To account for heterogeneity in AIV surveillance and minimise the risk of sampling pseudo-absences in poorly monitored regions, we restricted our analysis to countries (or administrative level 1 units in China and Russia) with at least five confirmed outbreaks.”

The reporting of spatial and temporal resolution of data in the manuscript could be significantly clearer. Is there a reason why human population density is downscaled to 5 arcminutes (~10km at the equator) while environmental covariate data has a resolution of 1km? The projection used is not reported. The authors should clarify the time period/resolution of the covariate data assigned to the occurrence dataset, for example, does "day LST annual mean" represent a particular year pre- or post-2020? Or an average over a number of years? Given that disease detections are associated with observation and reporting dates, and that there may be seasonal patterns in HPAI occurrence, it would be helpful to the reader to include this information when the eco-climatic indices are described. It would also be helpful to the reader to summarise the source, spatial and temporal resolution of all covariates in a table, as in Dhingra et al. Could the Authors clarify whether the duck density layer is farmed ducks or wild ducks?

The projection is WGS 84 (EPSG:4326) and the resolution of the output maps is around 0.0833 x 0.0833 decimal degrees (i.e. 5 arcmin, or approximately 10 km at the equator). We have now added these specifications in the text: “All maps are in a WGS84 projection with a spatial resolution of 0.0833 decimal degrees (i.e. 5 arcmin, or approximately 10 km at the equator).” In addition, we have now specified in the text that duck refers to domestic duck for clarity.

Environmental variables retrieved for our analyses were here available as values averaged over distinct periods of time (for further detail see Supplementary Information Resources S1 — description and source of each environmental variable included in the original sets of variables — available at https://github.com/sdellicour/h5nx_risk_mapping). In future works, this would indeed be interesting to associate the occurrences to a specific season with the variables accordingly, specially for viruses such as HPAI which have been found correlated with seasons. However, we did not conduct this type of analysis in the present study, occurrences being here associated with averaged values of environmental data only.

In line 407, the authors state a number of pseudo-absence points used in modelling, relative to the number of presence points, without clear justification. Note that relative weights can be assigned to occurrence data in most ECN software (e.g., R package gbm), to allow many pseudo-absence points to be sampled to represent the full extent of probable surveillance effort and subsequently down-weighted.

We thank the Reviewer for this suggestion. We acknowledge that alternative approaches such as down-weighting pseudo-absence points could offer a certain degree of flexibility in representing surveillance effort. However, we opted for a fixed 1:3 ratio of pseudoabsences to presence points within each administrative unit to ensure a consistent and conservative

sampling distribution. This approach aimed to limit overrepresentation of pseudoabsences in areas with sparse presence data, while still reflecting areas of likely surveillance.

There are a number of typographical errors and phrasing issues in the manuscript. A nonexhaustive list is provided below.

- Line 21: *"its" should be "their"* - Line 25: *"HPAI cases"*

Modifications have been done.

- Line 63: *sentence beginning "However" is somewhat out of context - what is it (briefly) about recent outbreaks that challenge existing models?*

We have now edited that sentence as follows: “However, recent outbreaks raise questions about whether earlier ecological niche models still accurately predict the current distribution of areas ecologically suitable for the local circulation of HPAI H5 viruses.”

- Lines 71 and 390: *"AIV" is not defined in the text* - Line 73: *"do" ("are" and "what" are not capitalised)*

Modifications have been done.

- Line 115: *"predictability" should be "predictive capacity"*

We have now replaced “predictability” by “predictive performance”.

- Line 180: *omit "pinpointing"*

- Line 192 *sentence beginning "In India," should be re-worded: is the point that there are detections of HPAI here and the model predicts high ecological suitability?*

- Line 195 *sentence beginning "Finally," phrasing could be clearer: Dhingra et al. find high suitability areas for H5Nx in North America which are predicted to be low suitability in the new model.*

- Line 237: *omit "the" in "with the those"*

- Line 374: *missing "."*

- Line 375: *"and" should be "to" (the same goes for line 421)*

- Line 448: *Rephrase "Simpson index goes" to "The Simpson index ranges"*

Modifications have been done.

Reviewer #2 (Public Review):

What is the justification for separating the dataset at 2020? Is it just the gap in-between the avian influenza outbreaks?

We chose 2020 as a cut-off based on a well-documented shift in HPAI epidemiology, notably the emergence and global spread of clade 2.3.4.4b, which may affect host dynamics and geographic patterns. We have now added this precision in the Materials and Methods section: “We selected 2020 as a cut-off point to reflect a well-documented shift in HPAI epidemiology, notably the emergence and global spread of clade 2.3.4.4b. This event marked a turning point in viral dynamics, influencing both the range of susceptible hosts and the geographical distribution of outbreaks.”

If the analysis aims to look at changing case numbers and distribution over time, surely the covariate datasets should be contemporaneous with the response?

Thank you for raising this important point. While we acknowledge that, ideally, covariates should match the response temporally, such high-resolution spatiotemporal environmental data were not available for most environmental factors considered in our ecological niche modelling analyses. While we used predictors (e.g., land-use variables, poultry density) that reflect long-term ecological suitability, we acknowledge that rather considering short-term seasonal variation could be an interesting perspective in future works, which is now explicitly stated in the Discussion section: “In addition, aligning outbreak occurrences with seasonally matched environmental variables could further refine predictions of HPAI risk linked to migratory dynamics.”

I would expect quite different immunity dynamics between domestic and wild birds as a function of lifespan and birth rates - though no obvious sign of that in the raw data. A statement on assumptions in that respect would be good.

Thank you for the comment. We agree that domestic and wild birds likely exhibit different immunity dynamics due to differences in lifespan, turnover rates, and exposure. However, our analyses did not explicitly model immunity processes, and the data did not show a clear signal of these differences.

Decisions and analytical tactics from Dhingra et al are adopted here in a way that doesn't quite convey the rationale, or justify its use here.

We thank the Reviewer for this observation. However, we do not agree with the notion that the rationale for using Dhingra et al.'s analytical framework is insufficiently conveyed. We adapted key components of their ecological niche modelling approach — such as the use of a boosted regression tree methodology and pseudo-absences sampling procedure — to ensure comparability with their previous findings, while also extending the analysis to additional time periods and host categories (wild vs. domestic birds). This framework aligns with the main objective of our study, which is to assess shifts in ecological suitability for HPAI over time and across host species, in light of changing viral dynamics.

Please go over the manuscript and harmonise the language about the model target - it is usually referred to as cases, but sometimes the pathogen, and others the wild and domestic birds where the cases were discovered.

We agree and we have now modified the text to only use the “cases” or “occurrences” terminology when referring to the model inputs.

Is the reporting of your BRT implementation correct? The text suggests that only 10 trees were run per replicate (of which there were 10 per response (domestic/wild x H5N1 / H5Nx) x distinct covariate set), but this would suggest that the authors were scarcely benefiting from the 'boosting' part of the BRTs that allow them to accurately estimate curvilinear functions. As additional trees are added, they should still be improving the loss function, and dramatically so in the early stages. The authors seem heavily guided by Elith et al's excellent paper[1] explaining BRTs and the companion tutorial piece, but in that work, the recommended approach is to run an initial model with a relatively quick learning rate that achieves the best fit to the held-out data at somewhere over 1000 trees, and then to refine the model to that number of trees with a slower learning rate. If the authors did indeed run only 10 trees I think that should be explained.

For each model, we used the “gbm.step” function to fit boosted regression trees, initiating the process with 10 trees and allowing up to 10,000 trees in steps of 5. The optimal number of trees was automatically determined by minimising the cross-validated deviance, following the recommended approach of Elith and colleagues (2008, J. Anim. Ecol.). This setup allows the boosting algorithm to iteratively improve model performance while avoiding overfitting. These aspects are now further clarified in the Materials and Methods section: “All BRT analyses were run and averaged over 10 cross-validated replicates, with a tree complexity of 4, a learning rate of 0.01, a tolerance parameter of 0.001, and while considering 5 spatial folds. Each model was initiated with 10 trees, and additional trees were incrementally added (in steps of 5) up to a maximum of 10,000, with the optimal number selected based on cross-validation tests.”

I'm uncomfortable with the strong interpretation of changes in indices such as those for diversity in the case of bird species with detected cases of avian influenza, and the relative influence of covariates in the environmental niche models. In the former case, if surveillance effort is increasing it might be expected that more species will be found to be infected. In the latter, I'm just not convinced that these fundamentally correlative models can support the interpretation of changing epidemiology as asserted by authors. This strikes me as particularly problematic in light of static and in some cases anachronistic predictor sets.

We thank the Reviewer for drawing attention to how changes in surveillance intensity might influence our diversity estimates. We have now integrated a new analysis to evaluate the increase in the number of wild birds tested and discussed the potential impact of this increase on the comparison of the bird species diversity metrics presented in our study, which is now interpreted with more caution: “To evaluate whether the post-2020 increase in species diversity estimated for infected wild birds could result from an increase in the number of tests performed on wild birds, we compared European annual surveillance test counts (EFSA et al., 2025, 2019) before and after 2020 using a Wilcoxon rank-sum test. We relied on European data because it was readily accessible and offered standardised and systematically collected metrics across multiple years, making it suitable for a comparative analysis. Although borderline significant (p-value = 0.063), the Wilcoxon rank-sum test indeed highlighted a recent increase in the number of wild bird tests (on average >11,000/year pre-2020 and >22,000 post-2020), which indicates that the comparison of bird species diversity metrics should be interpreted with caution. However, such an increase in the number of tests conducted in the context of a passive surveillance framework would thus also be in line with an increase in the number of wild birds found dead and thus tested. Therefore, while the increase in the number of tests could indeed impact species diversity metrics such as the Shannon index, it can also reflect an absolute higher wild bird mortality in line with a broadened range of infected bird species.”

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